

# Banana River Aquatic Preserve

## SEACAR Water Quality Analysis

Last compiled on 09 April, 2026

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# Indicators

## Nutrients

### Total Nitrogen - Discrete

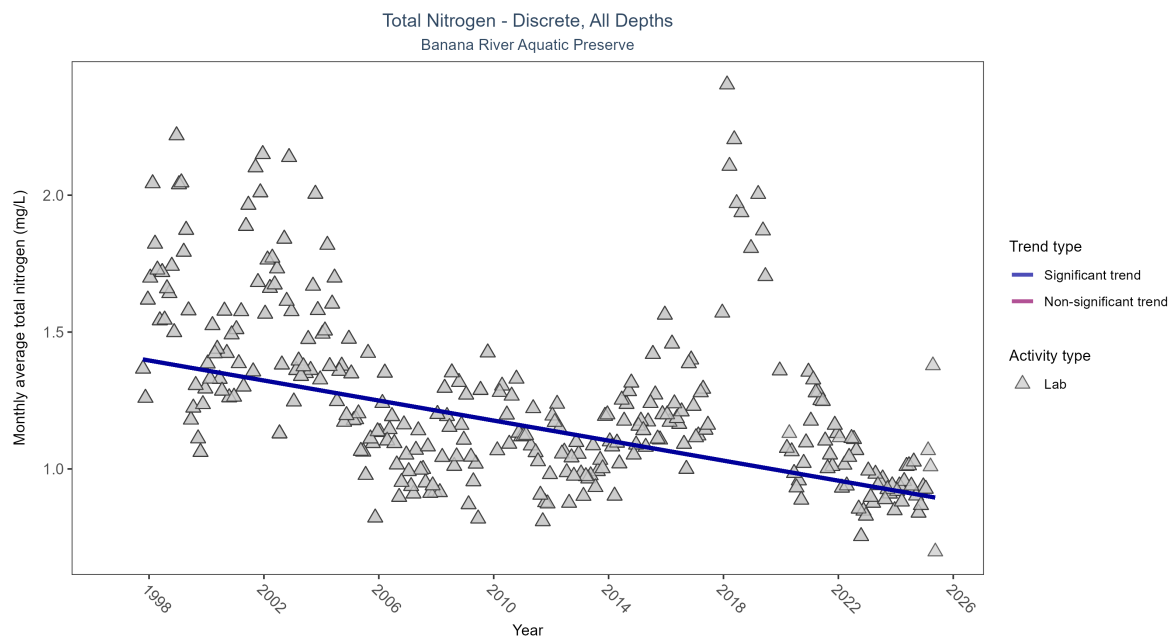


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|---|
| Lab           | Significantly decreasing trend | 2336         | 29              | 1997 - 2025      | 1.20646             | -0.41471 | 1.41447       | -0.01829  | 0 |

Monthly average total nitrogen decreased by 0.02 mg/L per year.

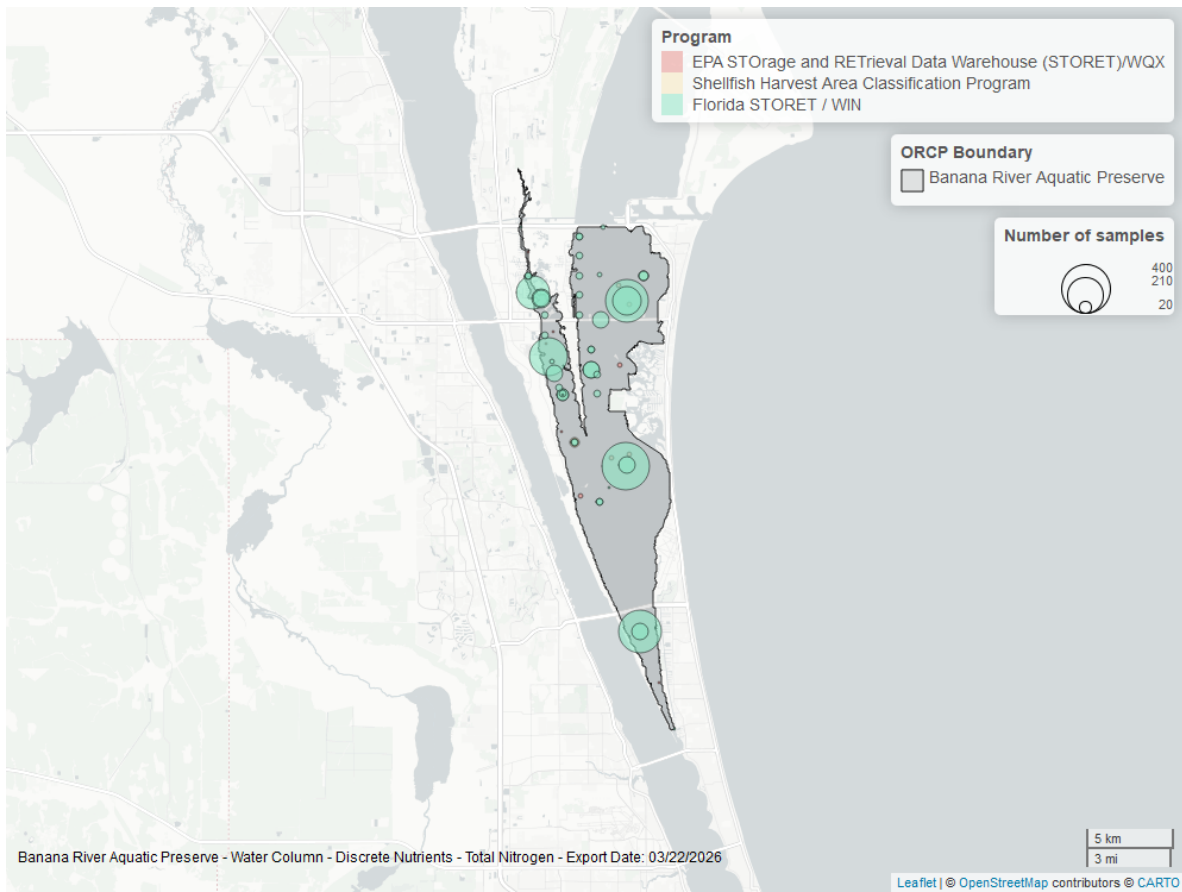


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Total Phosphorus - Discrete

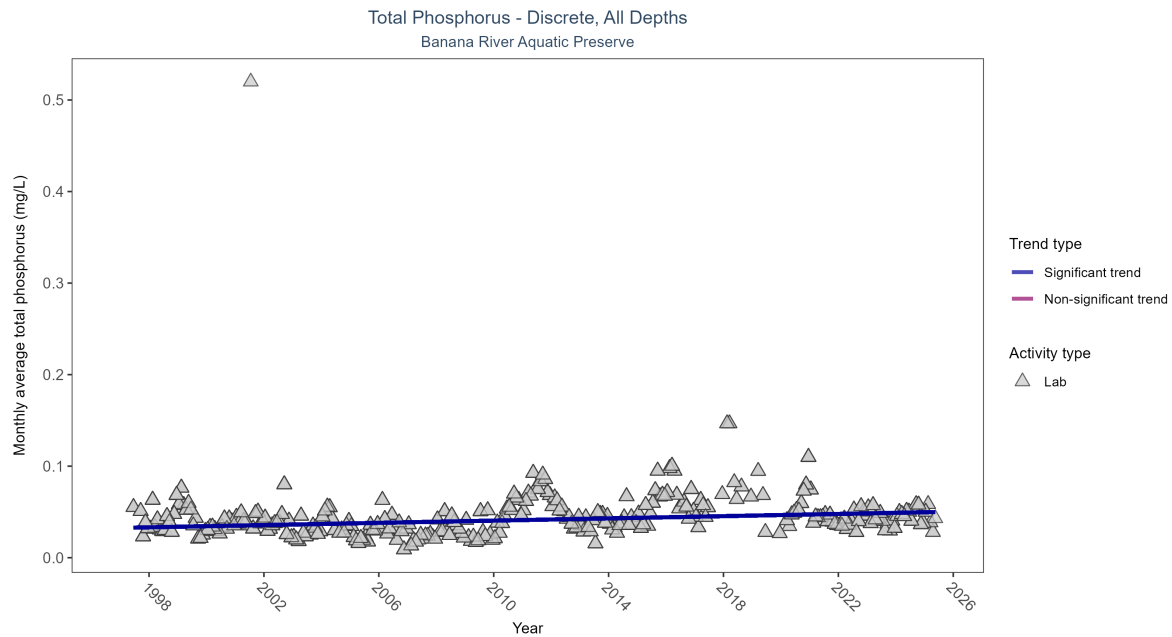


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|---|
| Lab           | Significantly increasing trend | 4827         | 29              | 1997 - 2025      | 0.0356              | 0.22415 | 0.03258       | 0.00061   | 0 |

Monthly average total phosphorus increased by less than 0.01 mg/L per year.



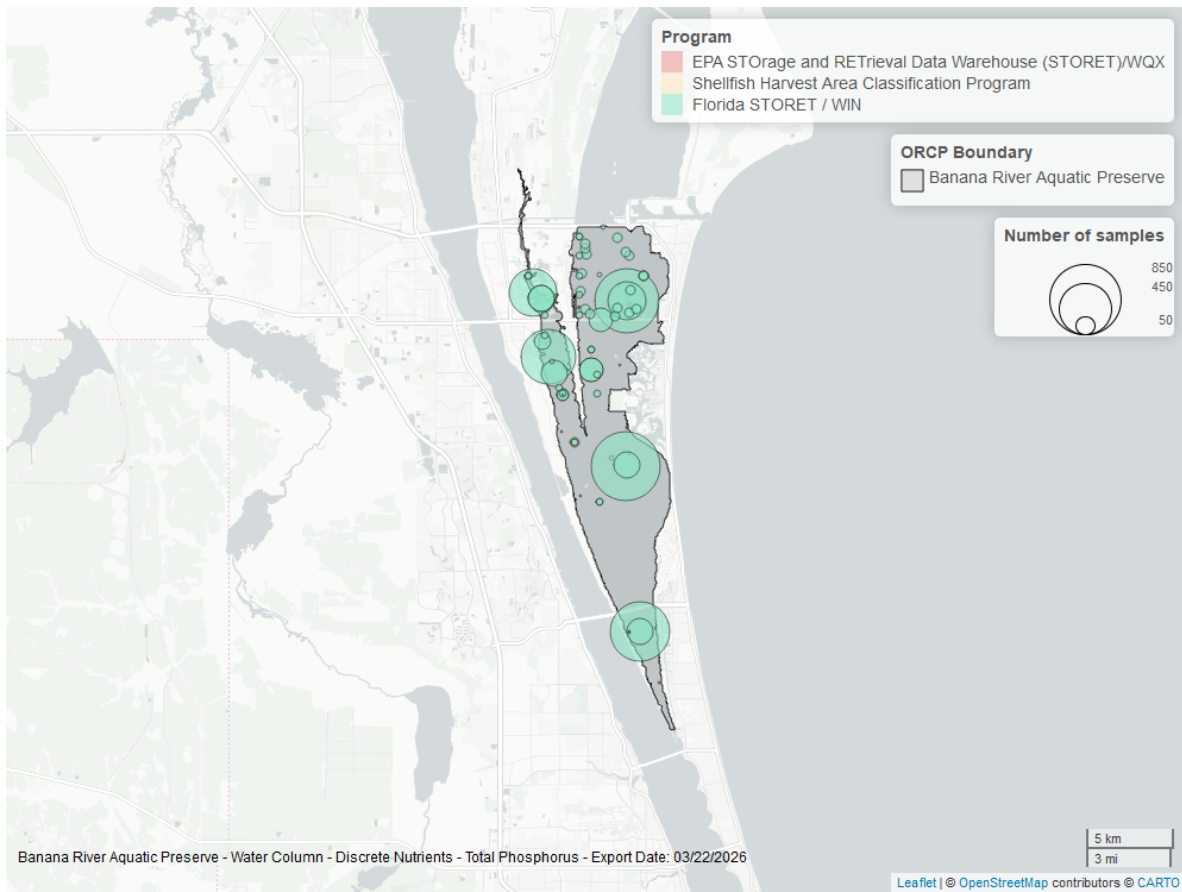


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Water Quality

### Dissolved Oxygen - Continuous

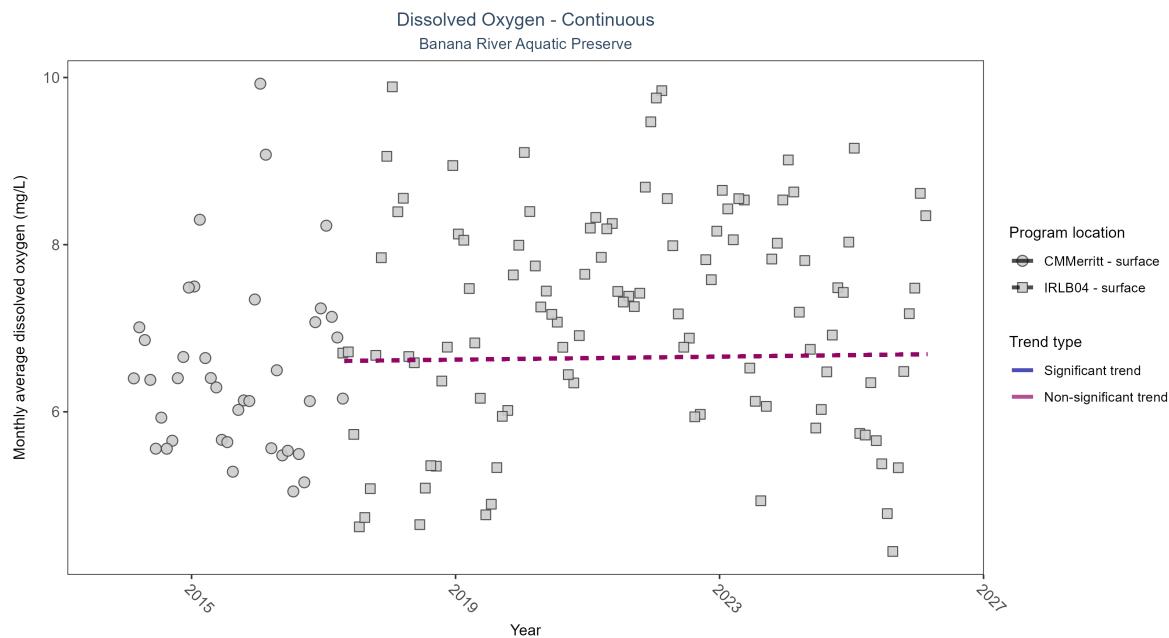


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: Seasonal Kendall-Tau Results - Dissolved Oxygen

| Program Location | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau  | Sen Intercept | Sen Slope | P      |
|------------------|--------------------------------------|--------------|-----------------|------------------|---------------------|------|---------------|-----------|--------|
| CMMerritt        | Insufficient data to calculate trend | 27378        | 4               | 2014 - 2017      | 6.51                | -    | -             | -         | -      |
| IRLB04           | No significant trend                 | 77153        | 10              | 2017 - 2026      | 7.21                | 0.03 | 6.61          | 0.01      | 0.6921 |

No detectable change in monthly average dissolved oxygen was observed at one location. There was insufficient data to fit a model for one location.

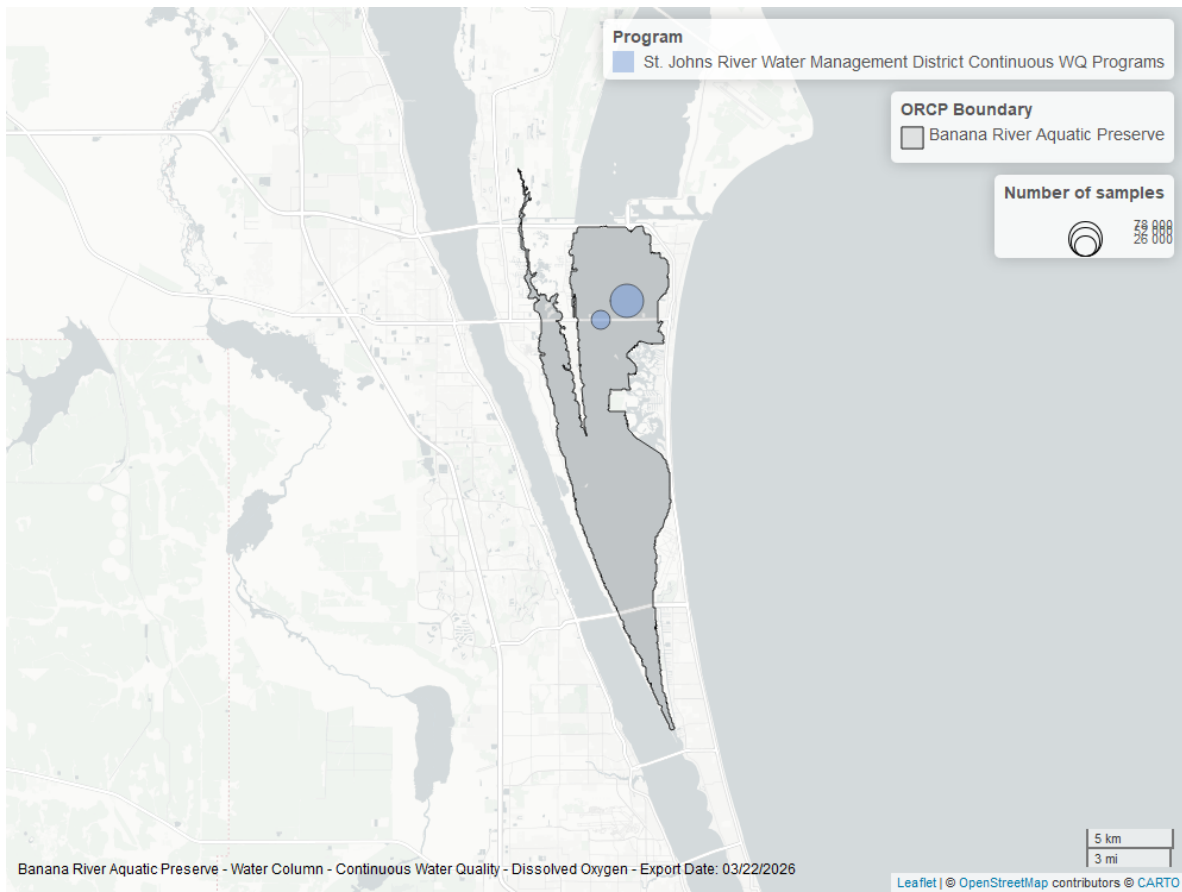


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Dissolved Oxygen - Discrete

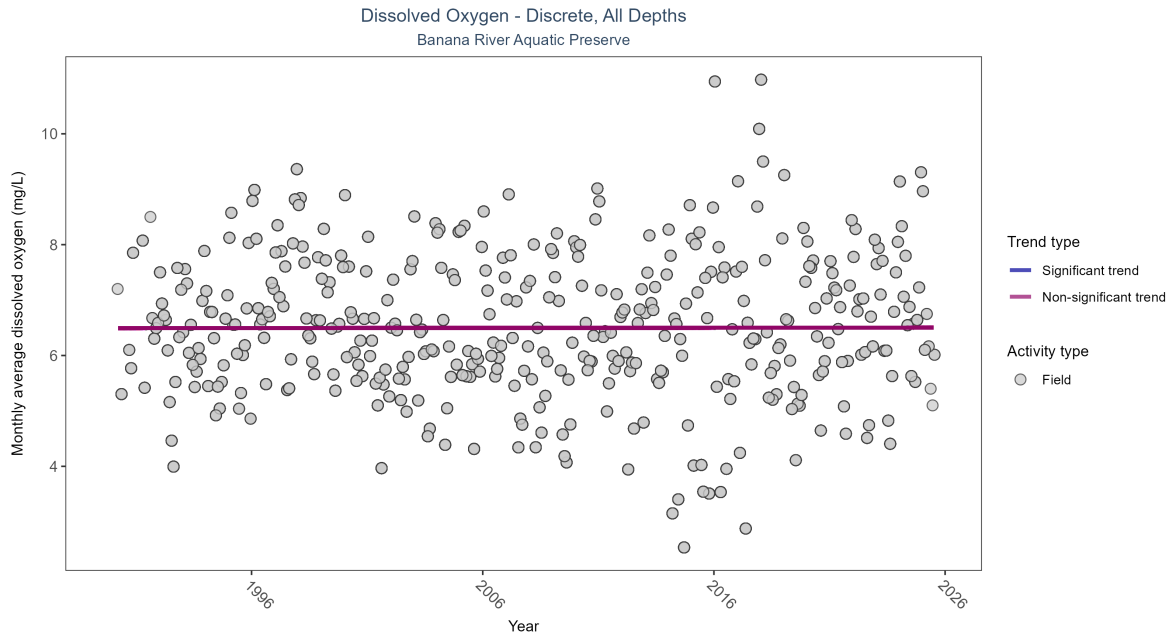


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen

| Activity Type | Statistical Trend    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau    | Sen Intercept | Sen Slope | P      |
|---------------|----------------------|--------------|-----------------|------------------|---------------------|--------|---------------|-----------|--------|
| Field         | No significant trend | 30082        | 36              | 1990 - 2025      | 6.5                 | 0.0049 | 6.49324       | 0.00027   | 0.9203 |

Dissolved oxygen showed no detectable trend between 1990 and 2025.

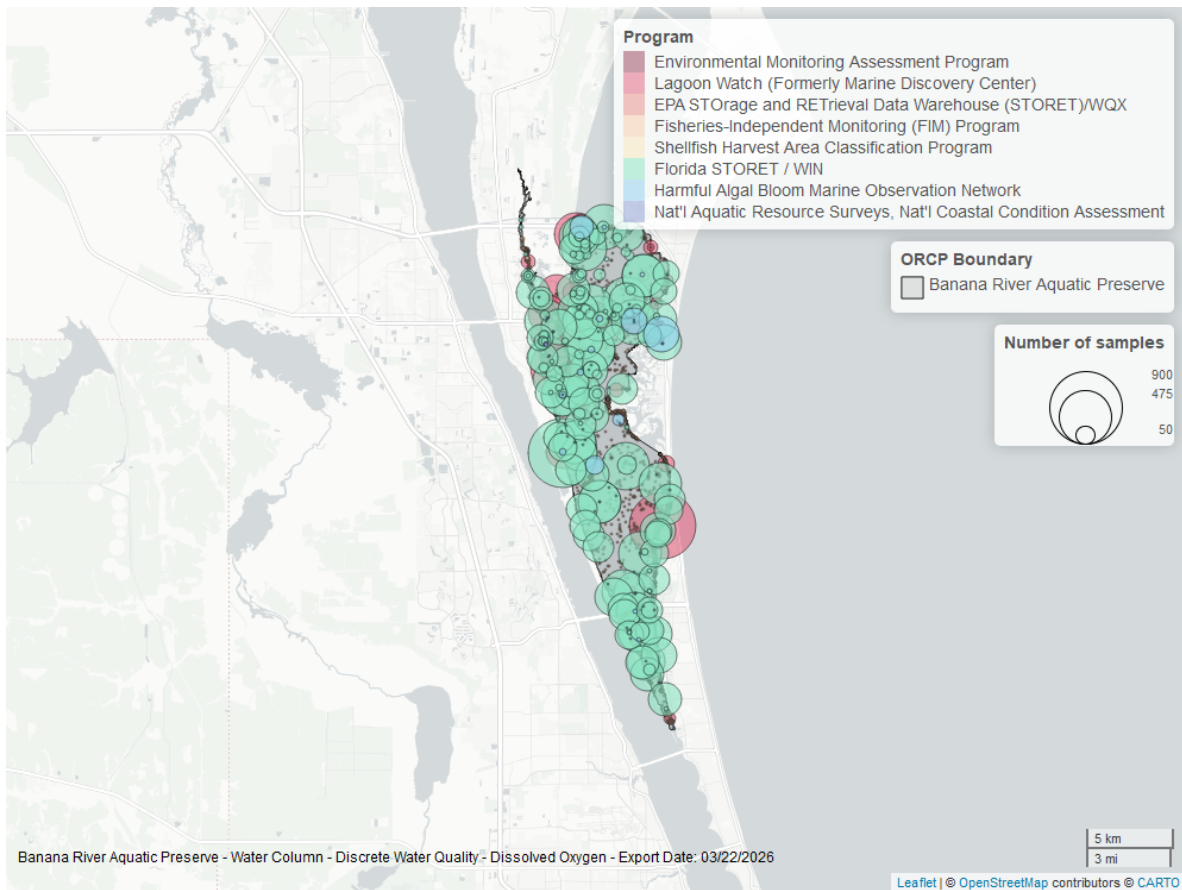


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Dissolved Oxygen Saturation - Continuous

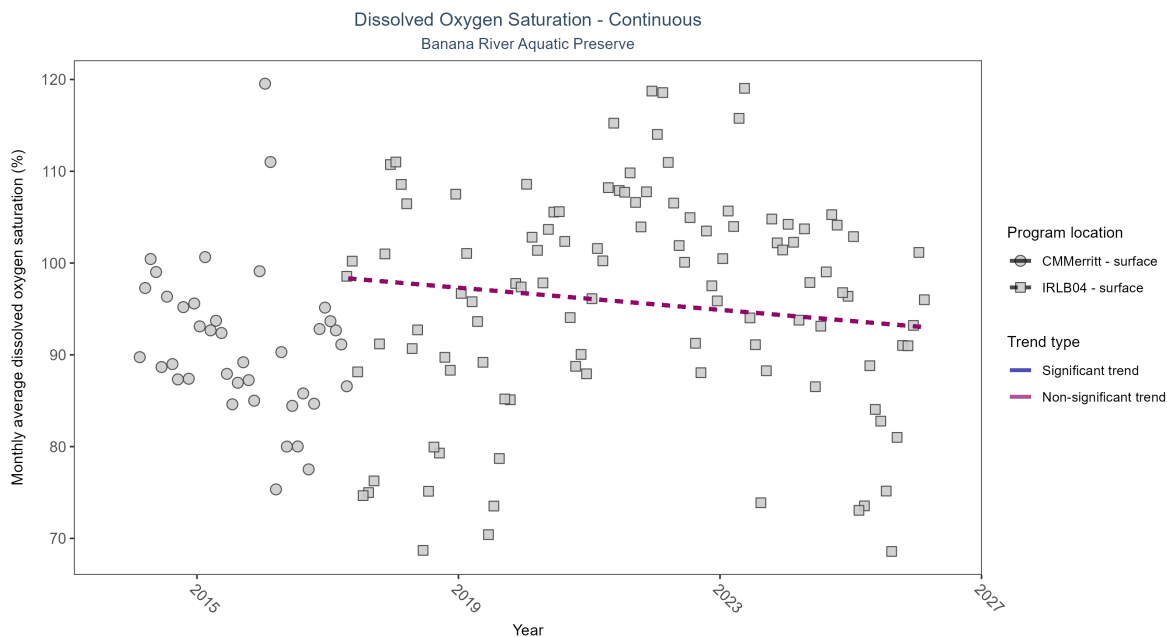


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: Seasonal Kendall-Tau Results - Dissolved Oxygen Saturation

| Program Location | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau   | Sen Intercept | Sen Slope | P      |
|------------------|--------------------------------------|--------------|-----------------|------------------|---------------------|-------|---------------|-----------|--------|
| CMMerritt        | Insufficient data to calculate trend | 25864        | 4               | 2014 - 2017      | 91.03               | -     | -             | -         | -      |
| IRLB04           | No significant trend                 | 89829        | 10              | 2017 - 2026      | 100.13              | -0.08 | 98.52         | -0.6      | 0.3147 |

No detectable change in monthly average dissolved oxygen saturation was observed at one location. There was insufficient data to fit a model for one location.

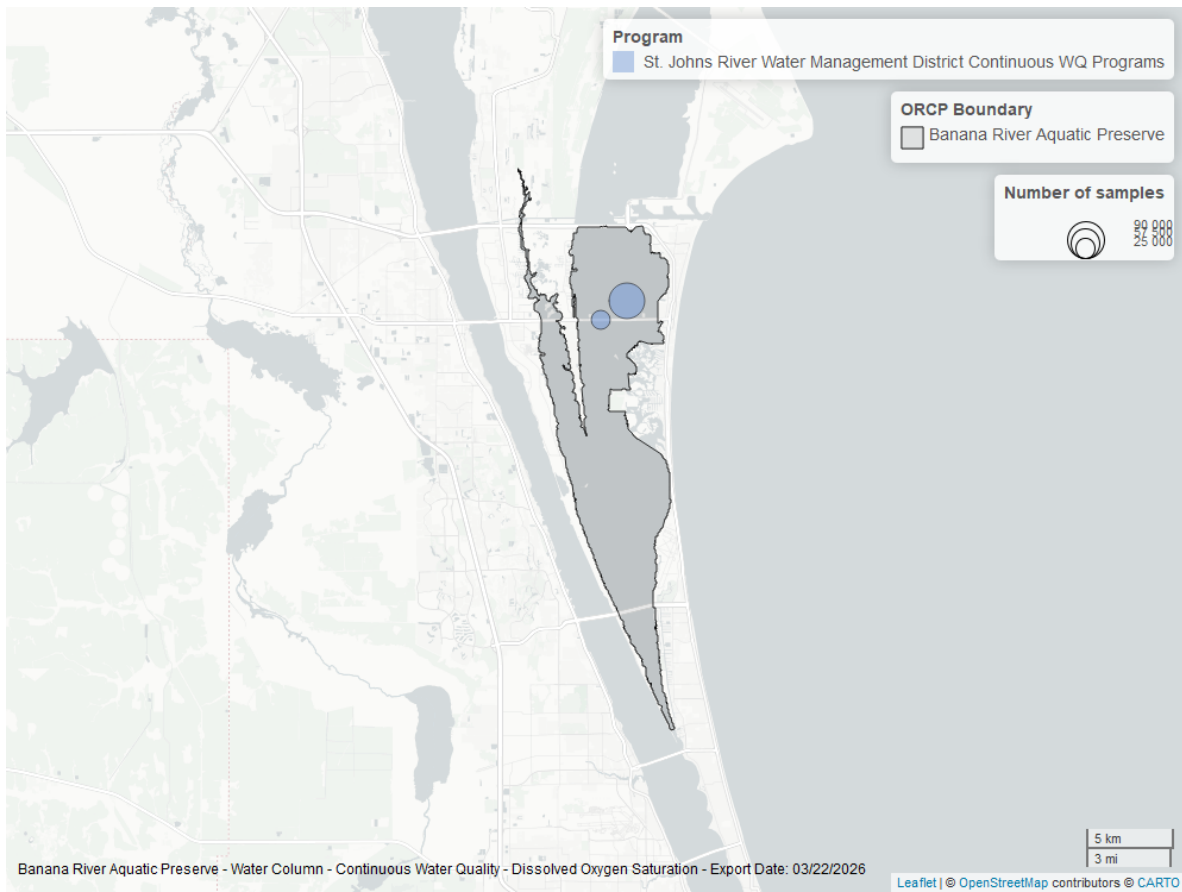


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Dissolved Oxygen Saturation - Discrete

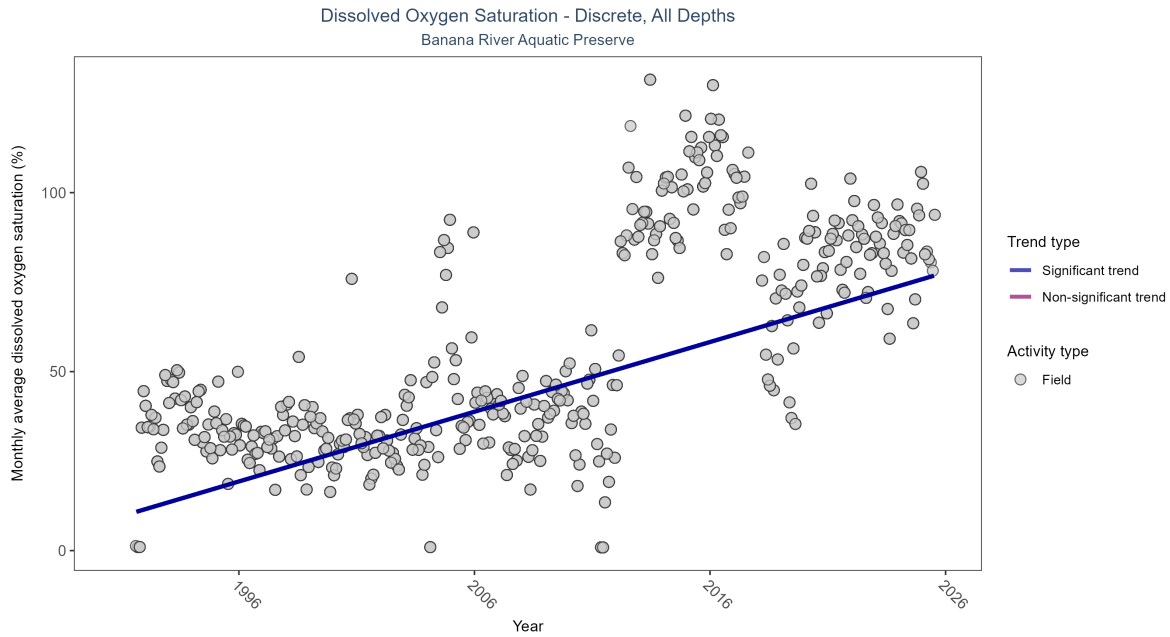


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|---|
| Field         | Significantly increasing trend | 7670         | 35              | 1991 - 2025      | 60                  | 0.46063 | 9.57314       | 1.94798   | 0 |

Monthly average dissolved oxygen saturation increased by 1.95% per year.



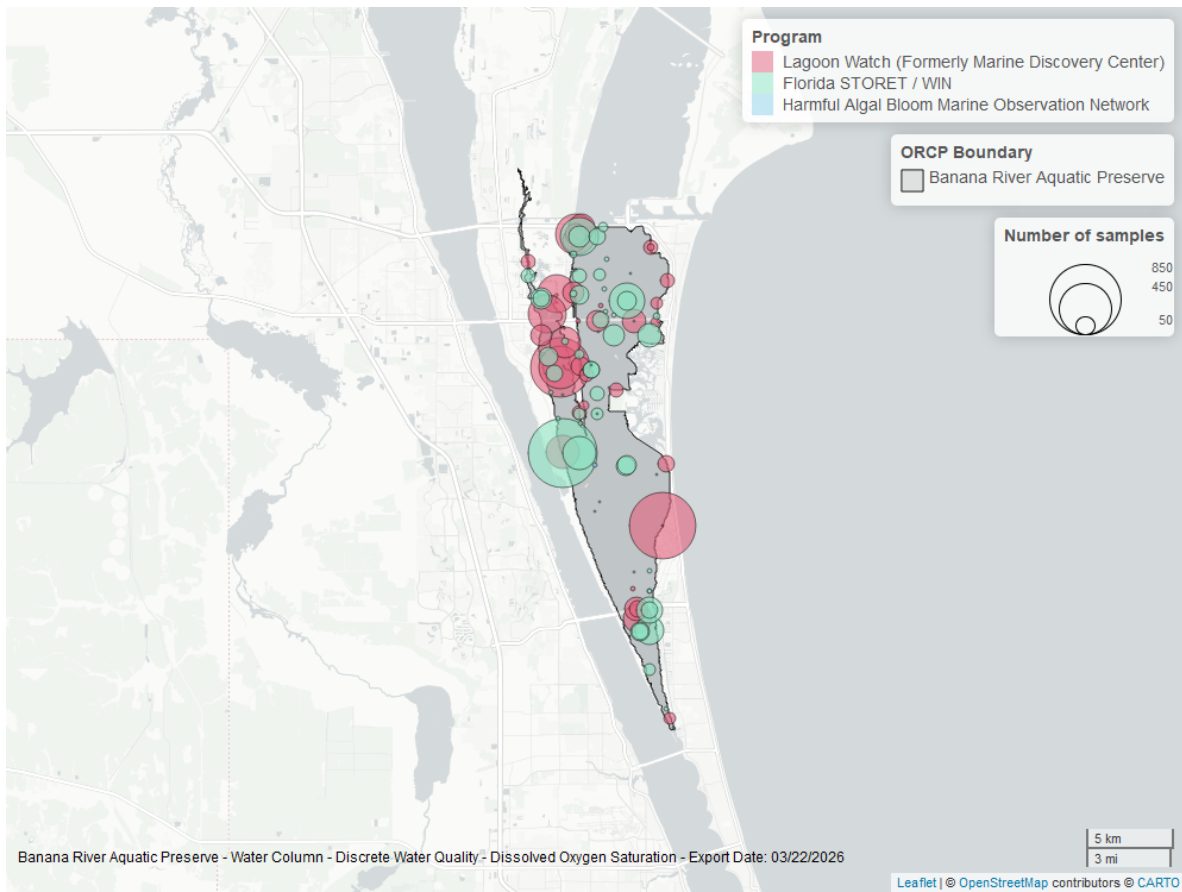


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Salinity - Discrete

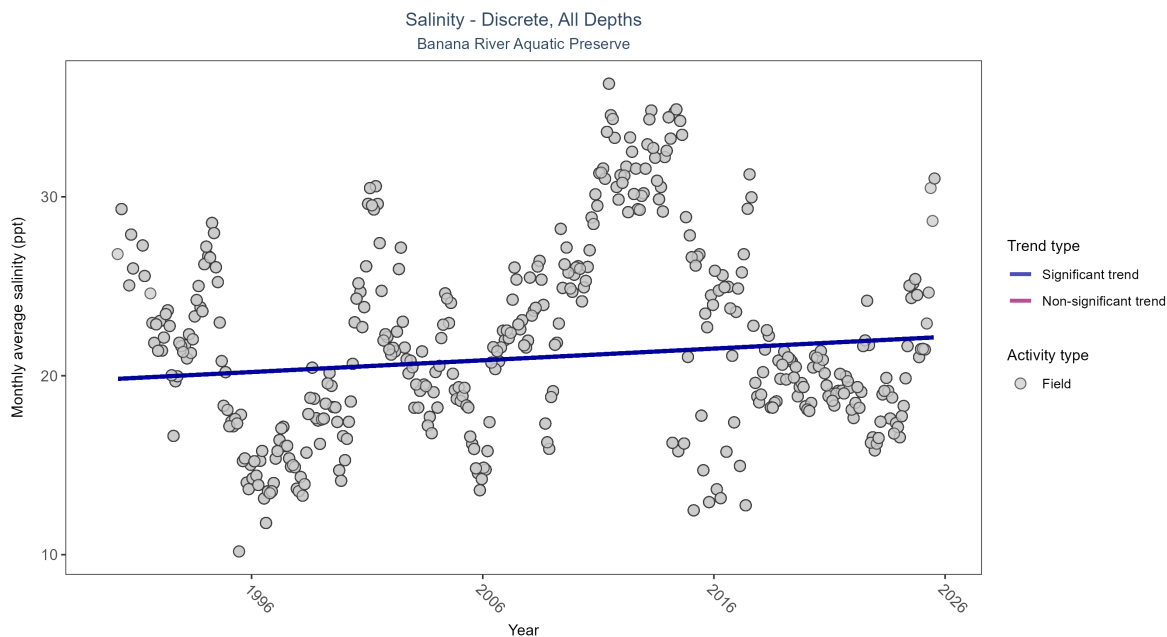


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Seasonal Kendall-Tau Results for - Salinity

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P      |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|--------|
| All           | Significantly increasing trend | 31776        | 36              | 1990 - 2025      | 19.8                | 0.08681 | 19.81342      | 0.06554   | 0.0123 |

Monthly average salinity increased by 0.07 ppt per year.

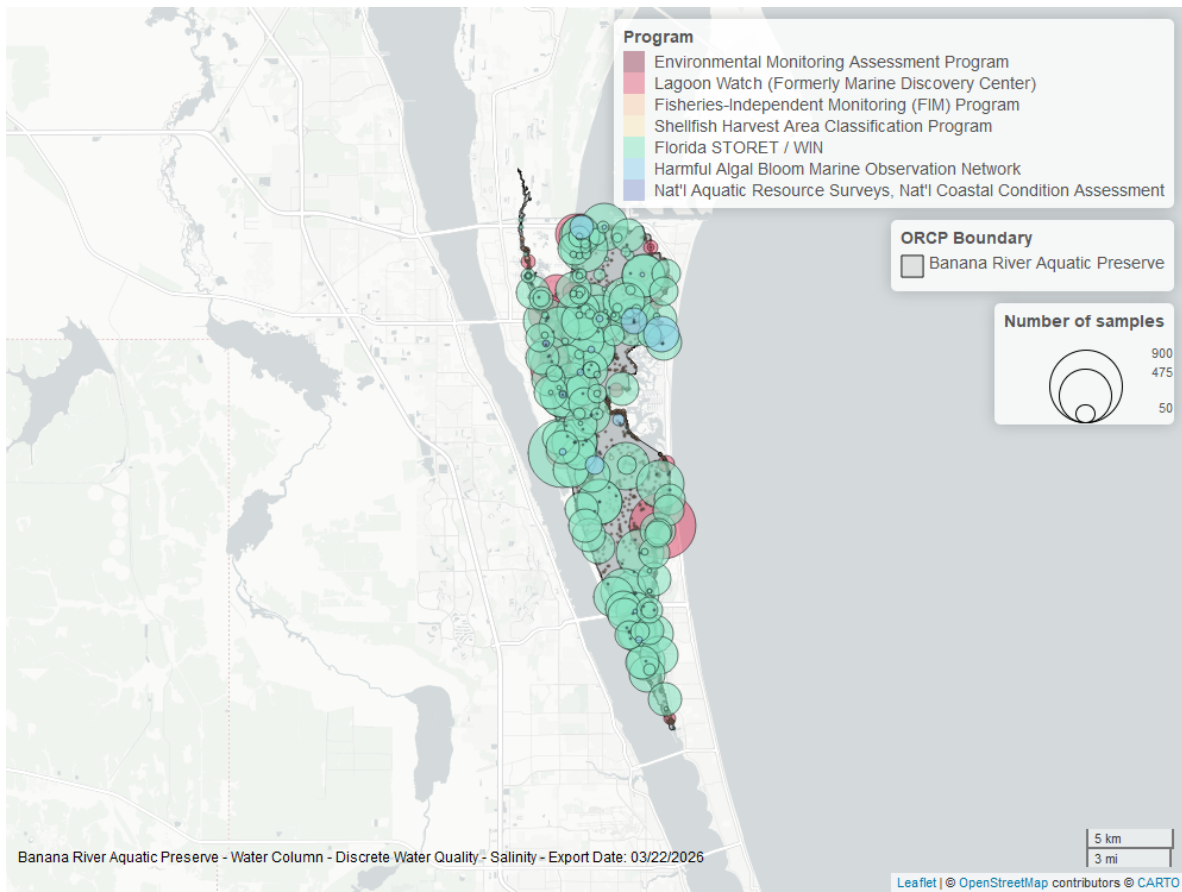


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Salinity - Continuous

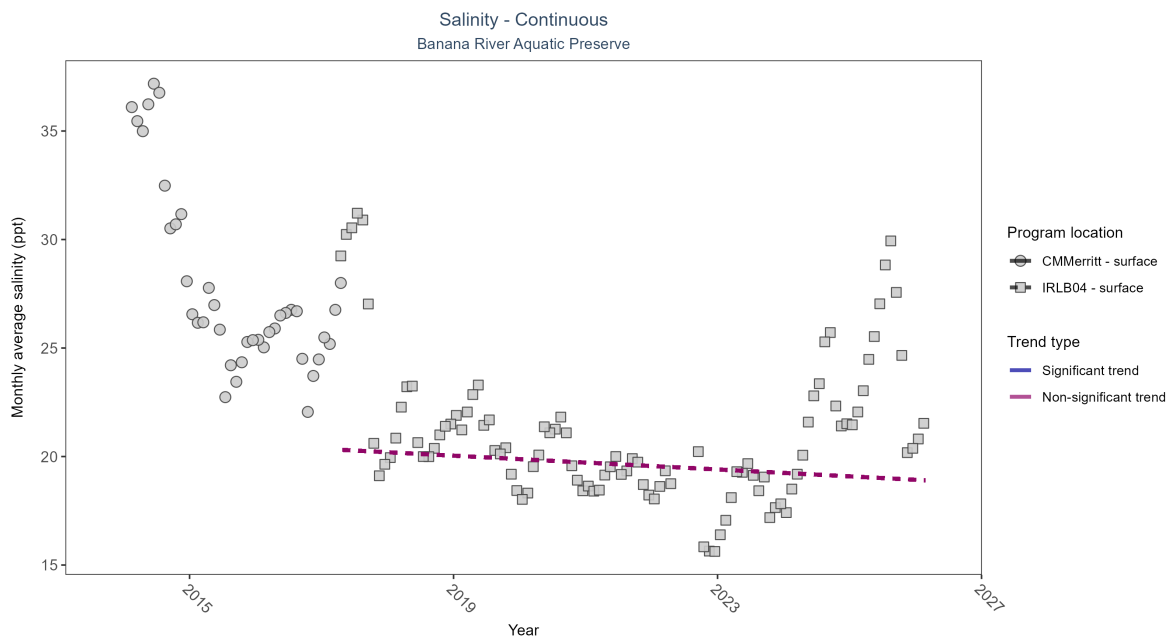


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: Seasonal Kendall-Tau Results - Salinity

| Program Location | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau  | Sen Intercept | Sen Slope | P      |
|------------------|--------------------------------------|--------------|-----------------|------------------|---------------------|------|---------------|-----------|--------|
| CMMerritt        | Insufficient data to calculate trend | 25902        | 4               | 2014 - 2017      | 26.40               | -    | -             | -         | -      |
| IRLB04           | No significant trend                 | 73363        | 10              | 2017 - 2026      | 20.28               | -0.1 | 20.36         | -0.16     | 0.2106 |

No detectable change in monthly average salinity was observed at one location. There was insufficient data to fit a model for one location.

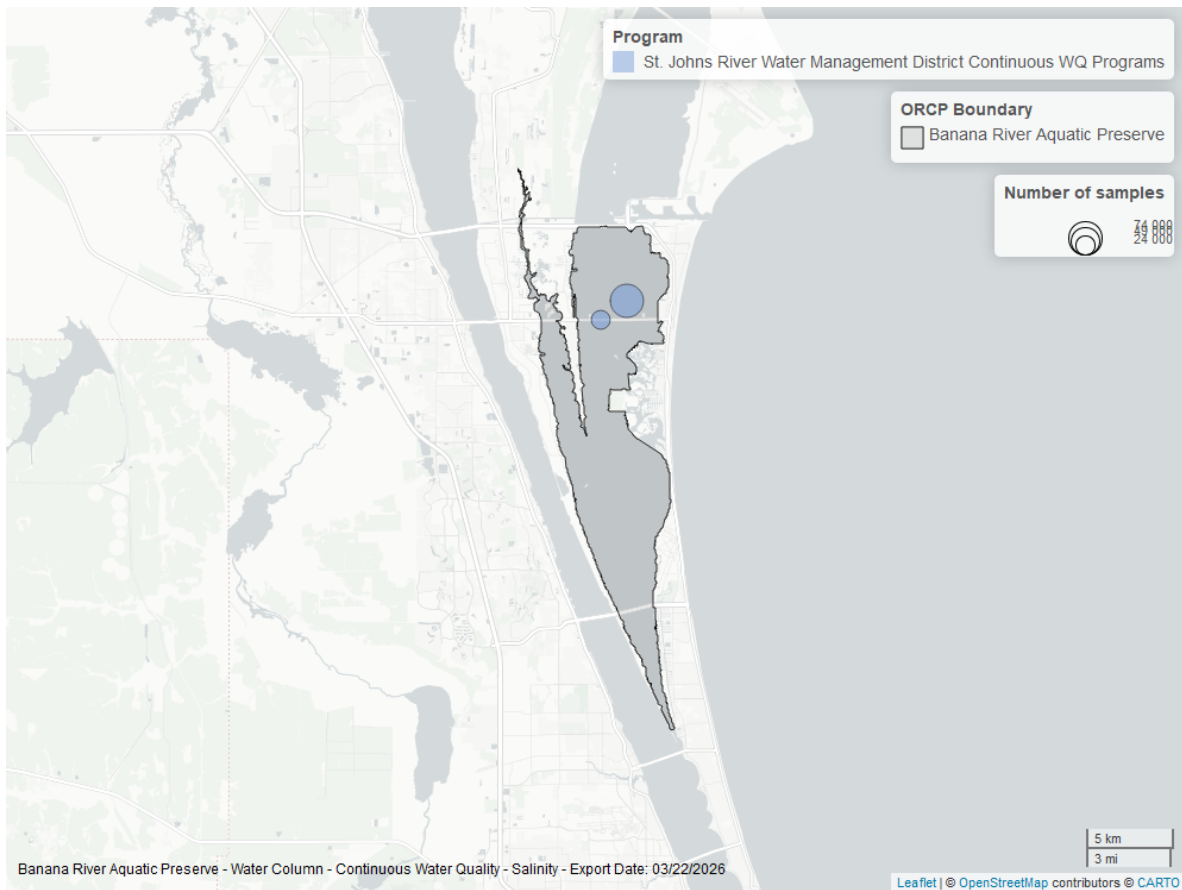


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Water Temperature - Continuous

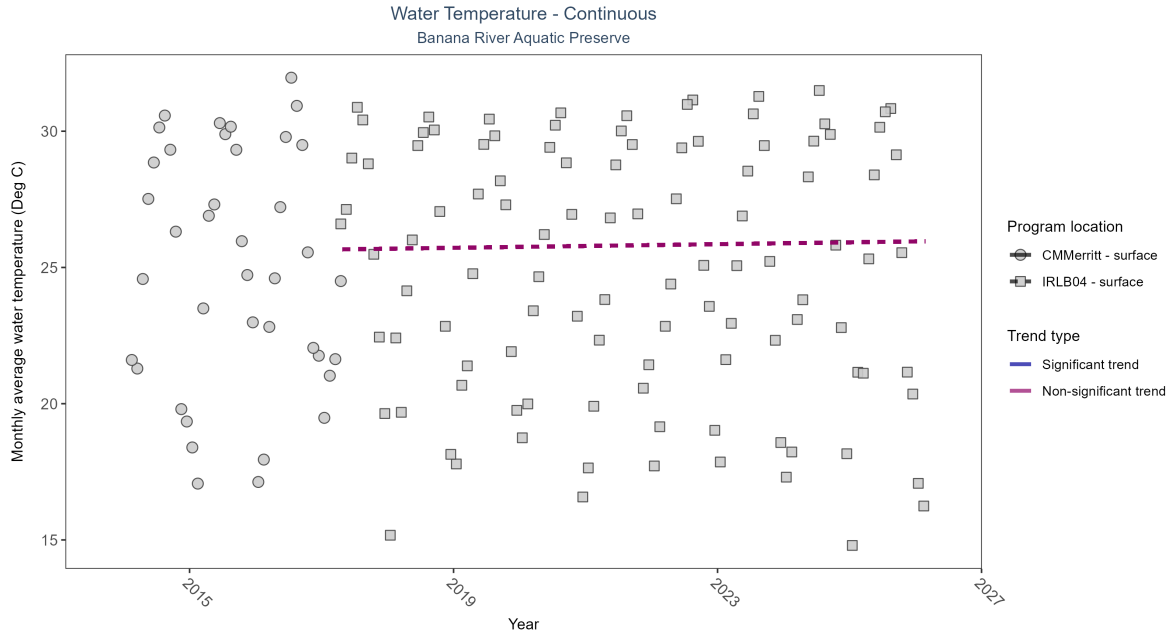


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: Seasonal Kendall-Tau Results - Water Temperature

| Program Location | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau  | Sen Intercept | Sen Slope | P      |
|------------------|--------------------------------------|--------------|-----------------|------------------|---------------------|------|---------------|-----------|--------|
| CMMerritt        | Insufficient data to calculate trend | 27484        | 4               | 2014 - 2017      | 25.43               | -    | -             | -         | -      |
| IRLB04           | No significant trend                 | 77157        | 10              | 2017 - 2026      | 25.61               | 0.05 | 25.65         | 0.03      | 0.5223 |

No detectable change in monthly average water temperature was observed at one location. There was insufficient data to fit a model for one location.

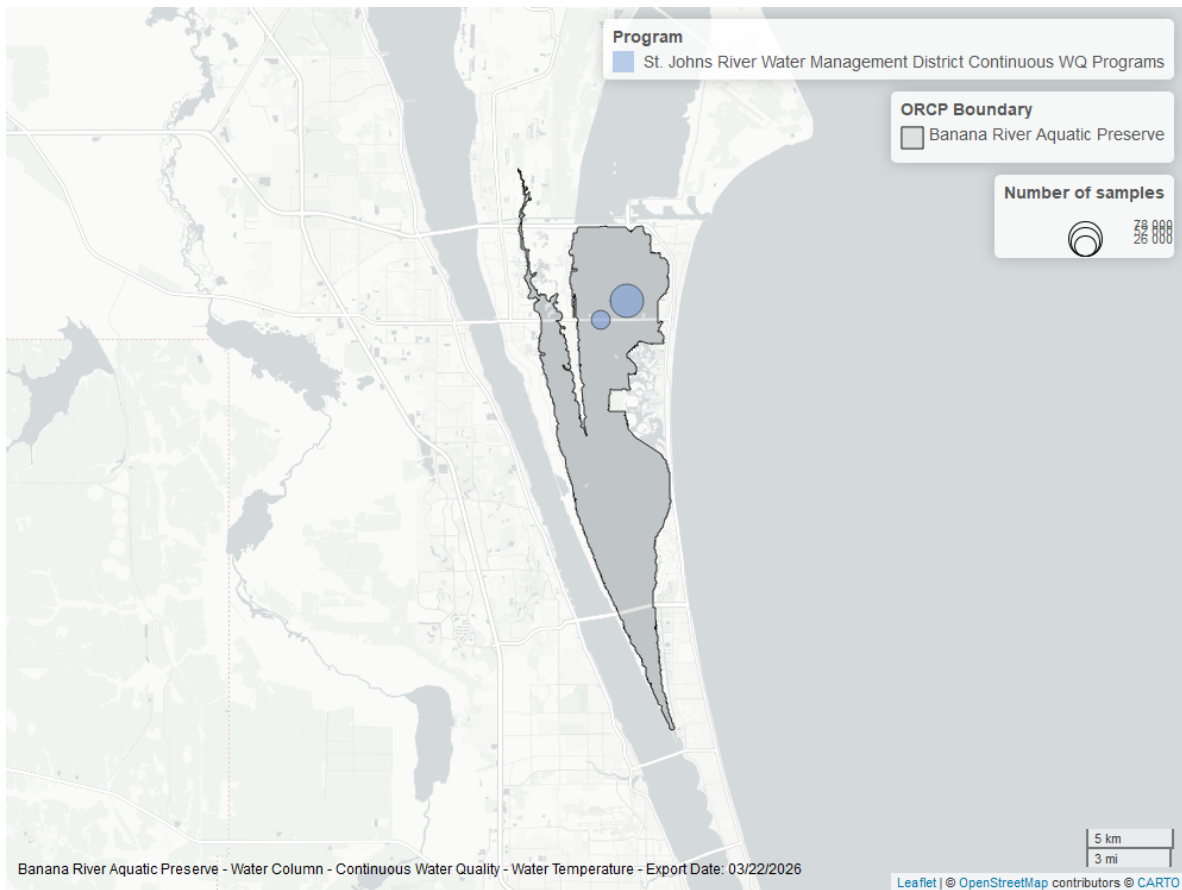


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Water Temperature - Discrete

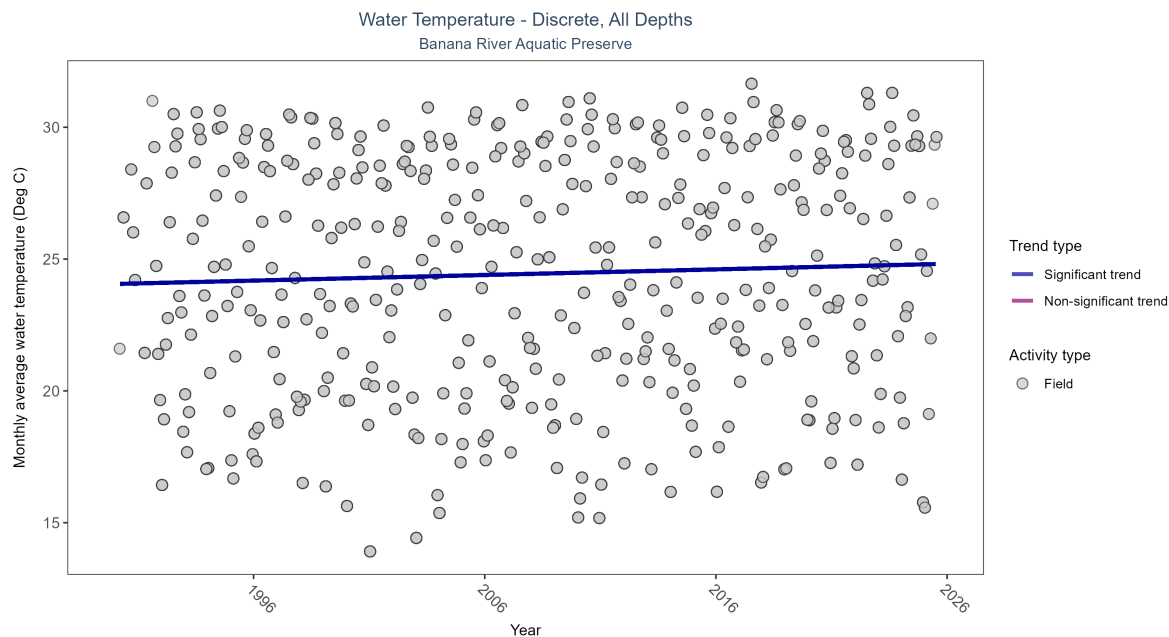


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Results for - Water Temperature

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P      |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|--------|
| Field         | Significantly increasing trend | 31615        | 36              | 1990 - 2025      | 25.5                | 0.12084 | 24.05574      | 0.02128   | 0.0004 |

Monthly average water temperature increased by 0.02°C per year.



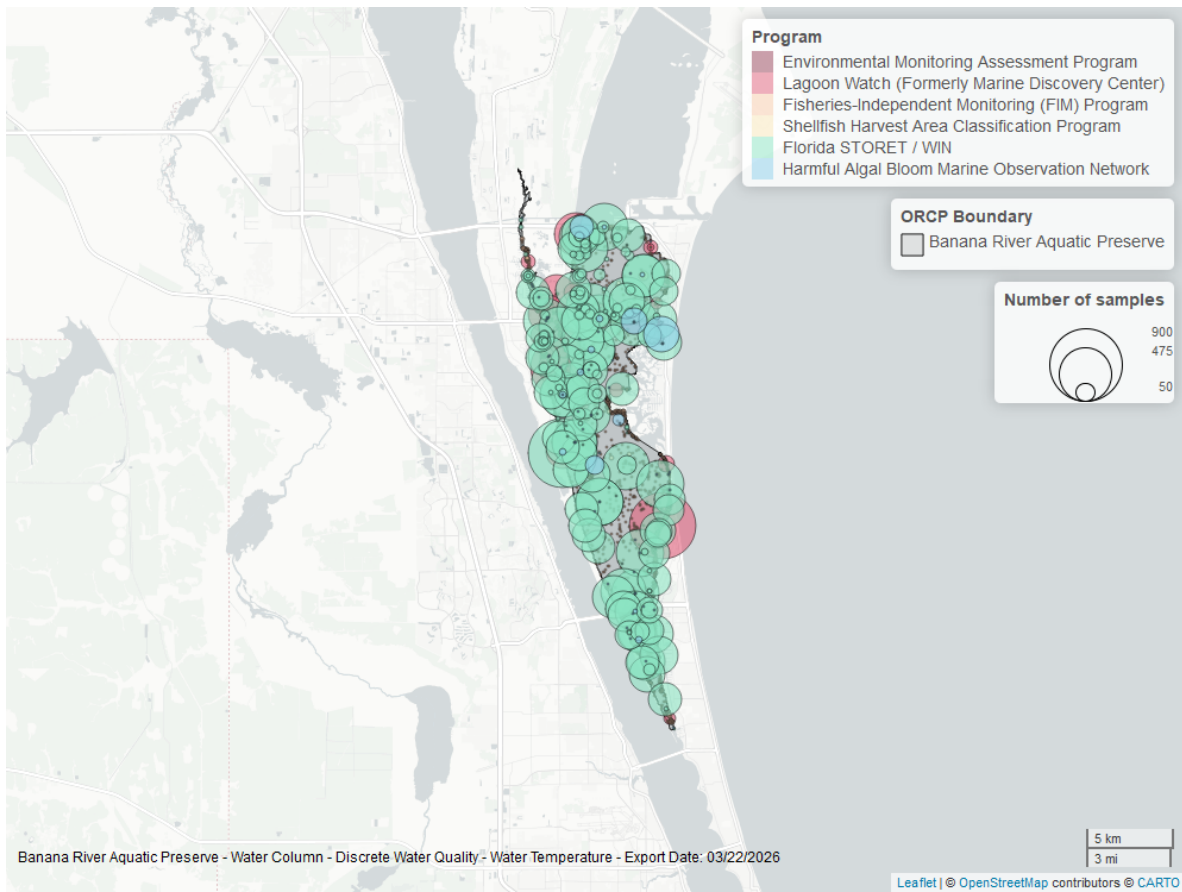


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## pH - Continuous

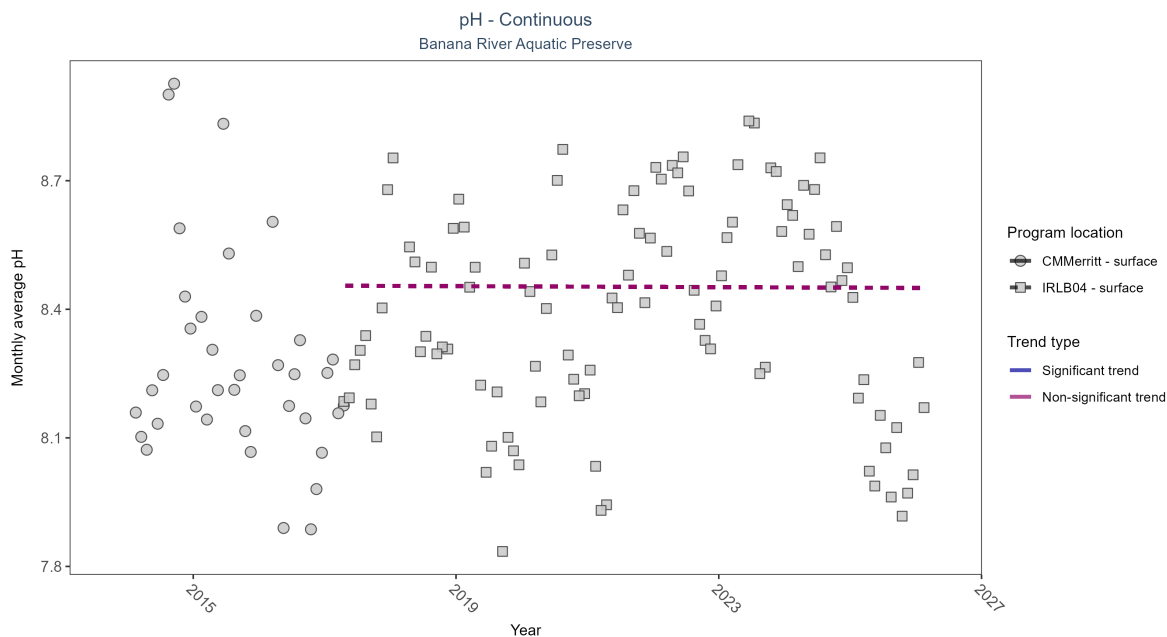


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: Seasonal Kendall-Tau Results - pH

| Program Location | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau   | Sen Intercept | Sen Slope | P      |
|------------------|--------------------------------------|--------------|-----------------|------------------|---------------------|-------|---------------|-----------|--------|
| CMMerritt        | Insufficient data to calculate trend | 27417        | 4               | 2014 - 2017      | 8.22                | -     | -             | -         | -      |
| IRLB04           | No significant trend                 | 77069        | 10              | 2017 - 2026      | 8.43                | -0.01 | 8.46          | 0         | 0.9272 |

No detectable change in monthly average pH was observed at one location. There was insufficient data to fit a model for one location.

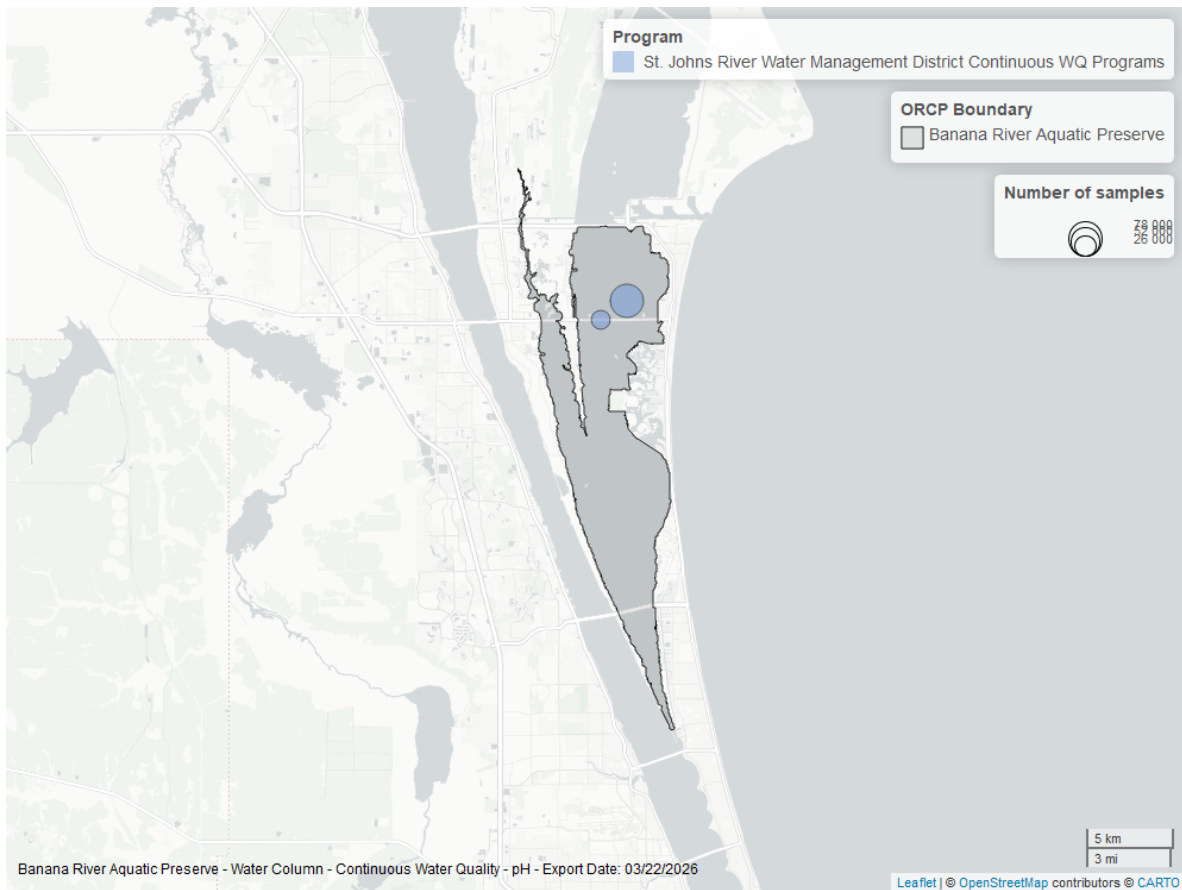


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## pH - Discrete

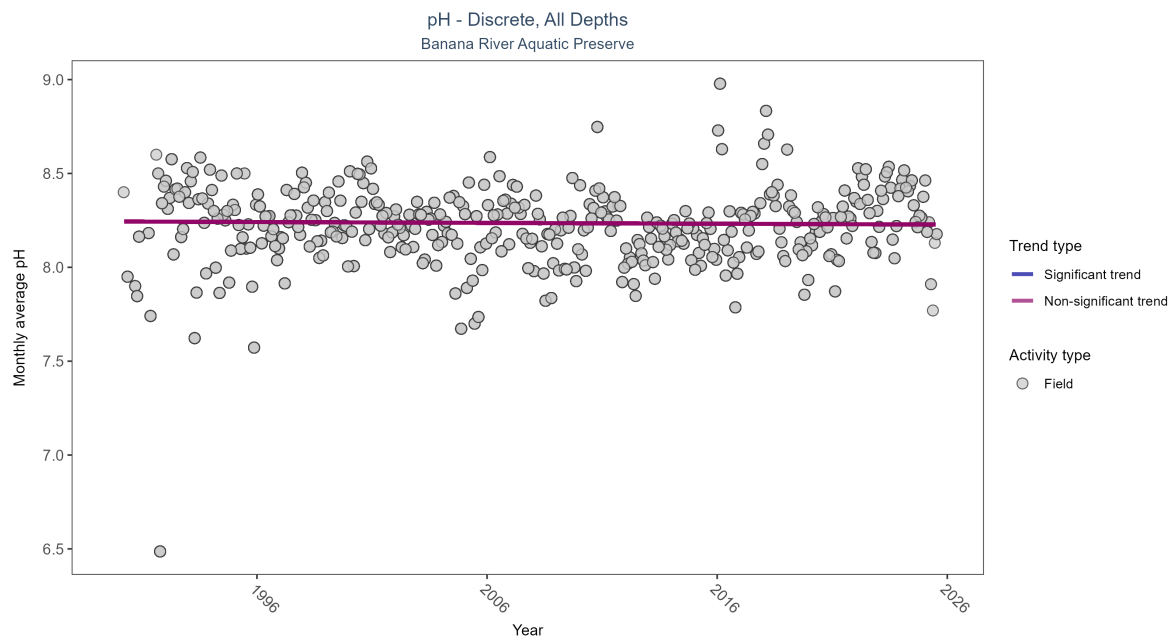


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Results for - pH

| Activity Type | Statistical Trend    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P      |
|---------------|----------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|--------|
| Field         | No significant trend | 23027        | 36              | 1990 - 2025      | 8.2                 | -0.01886 | 8.24397       | -0.00046  | 0.6285 |

pH showed no detectable trend between 1990 and 2025.

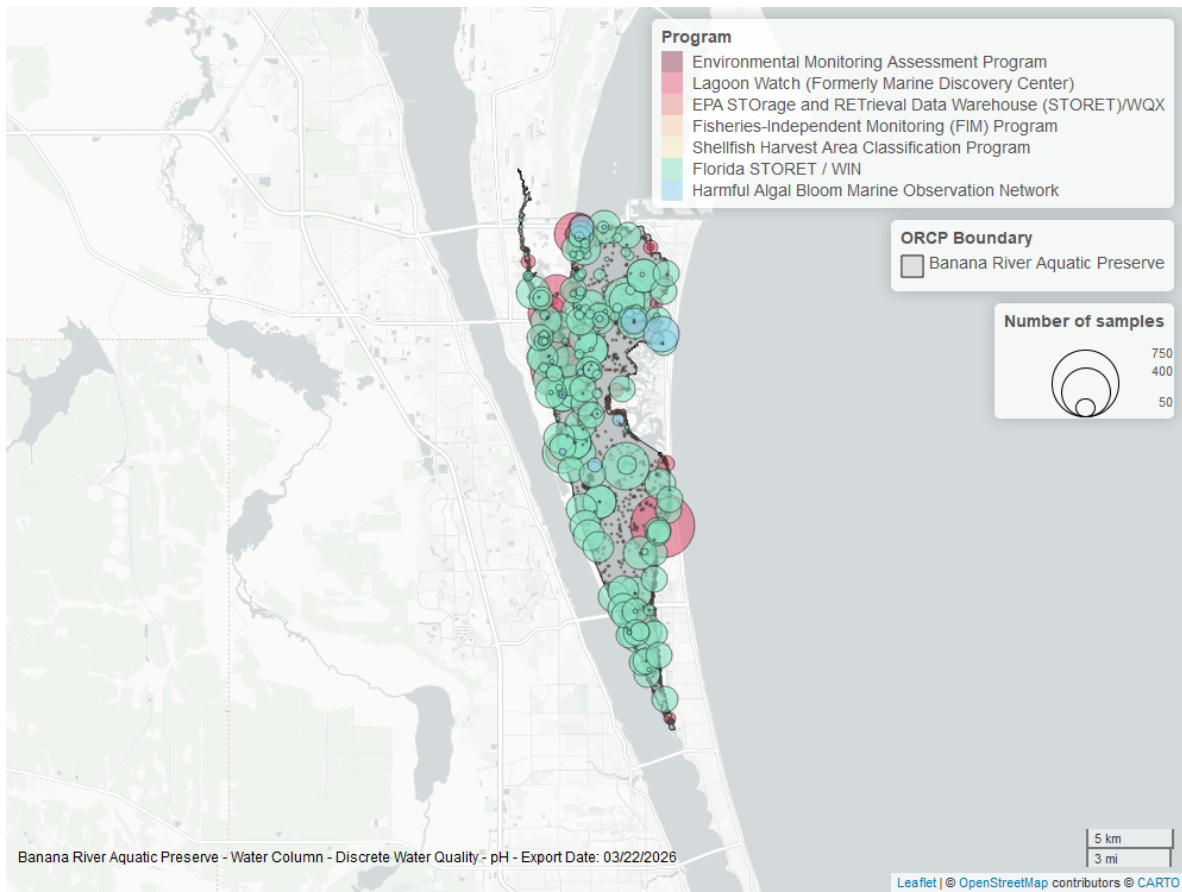


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Specific Conductivity - Continuous

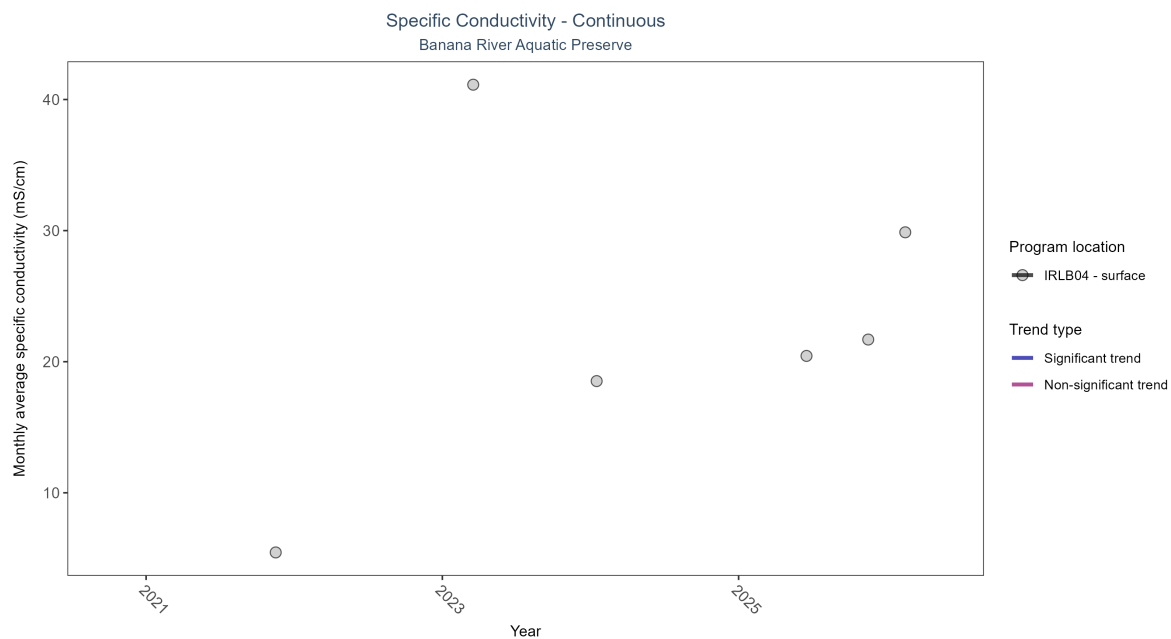


Table 13: Seasonal Kendall-Tau Results - Specific Conductivity

| Program Location | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|------------------|--------------------------------------|--------------|-----------------|------------------|---------------------|-----|---------------|-----------|---|
| IRLB04           | Insufficient data to calculate trend | 7            | 5               | 2021 - 2026      | 20.44               | -   | -             | -         | - |

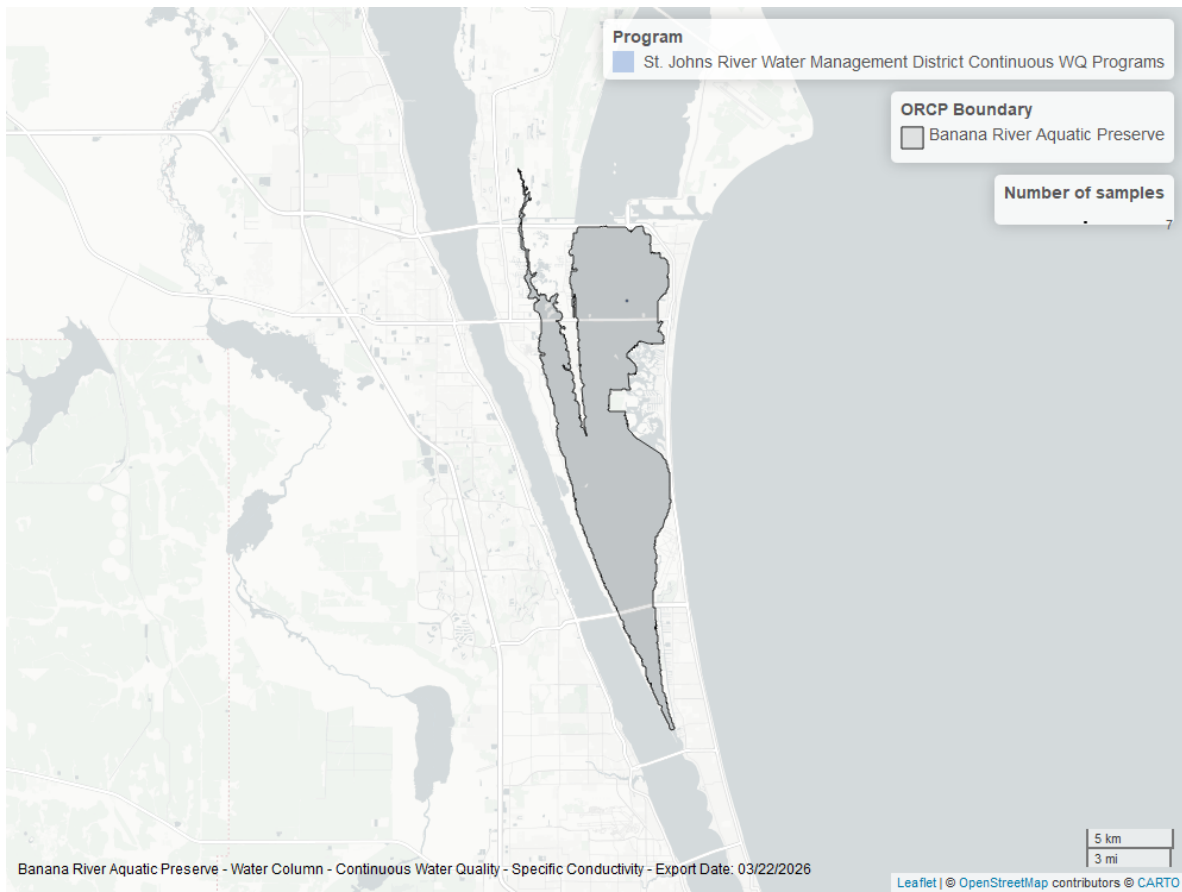


Figure 25: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

# Water Clarity

## Turbidity - Continuous

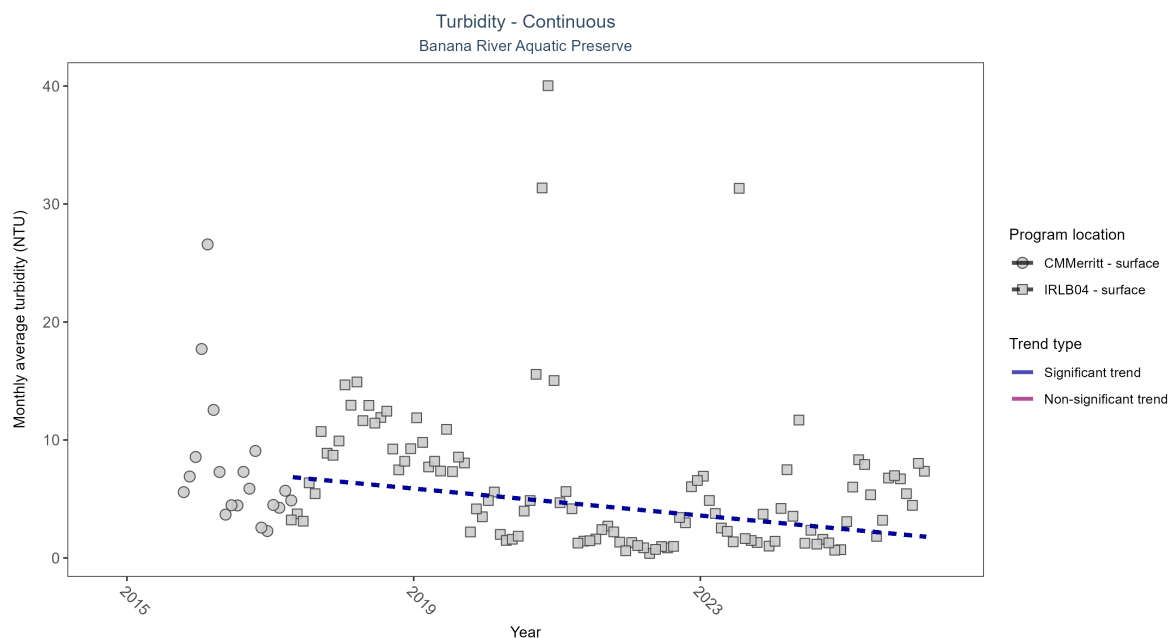


Figure 26: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: Seasonal Kendall-Tau Results - Turbidity

| Program Location | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau   | Sen Intercept | Sen Slope | P |
|------------------|--------------------------------------|--------------|-----------------|------------------|---------------------|-------|---------------|-----------|---|
| CMMerritt        | Insufficient data to calculate trend | 12912        | 3               | 2015 - 2017      | 5.29                | -     | -             | -         | - |
| IRLB04           | Significantly decreasing trend       | 74218        | 10              | 2017 - 2026      | 3.41                | -0.32 | 7.02          | -0.57     | 0 |

At one program location, monthly average turbidity decreased by 0.57 NTU per year. There was insufficient data to fit a model for one location.



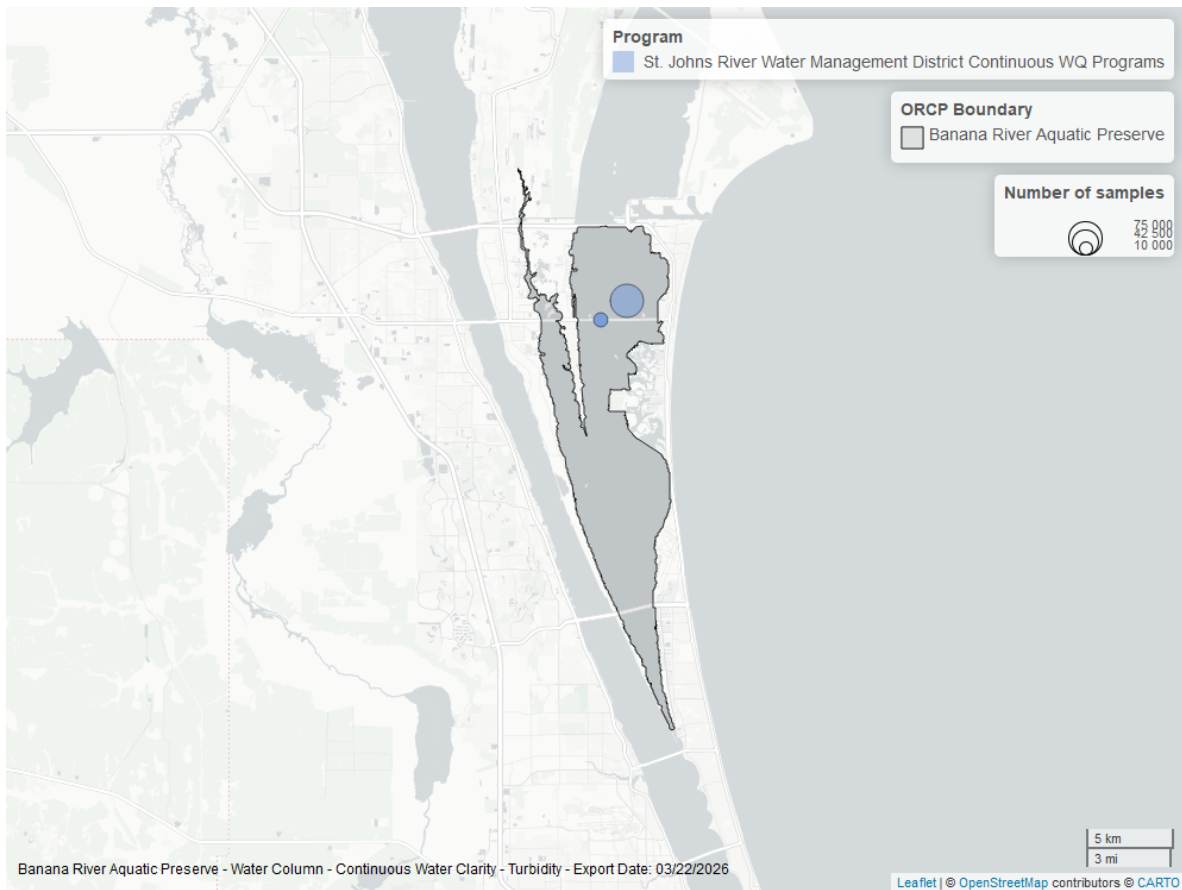


Figure 27: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Turbidity - Discrete

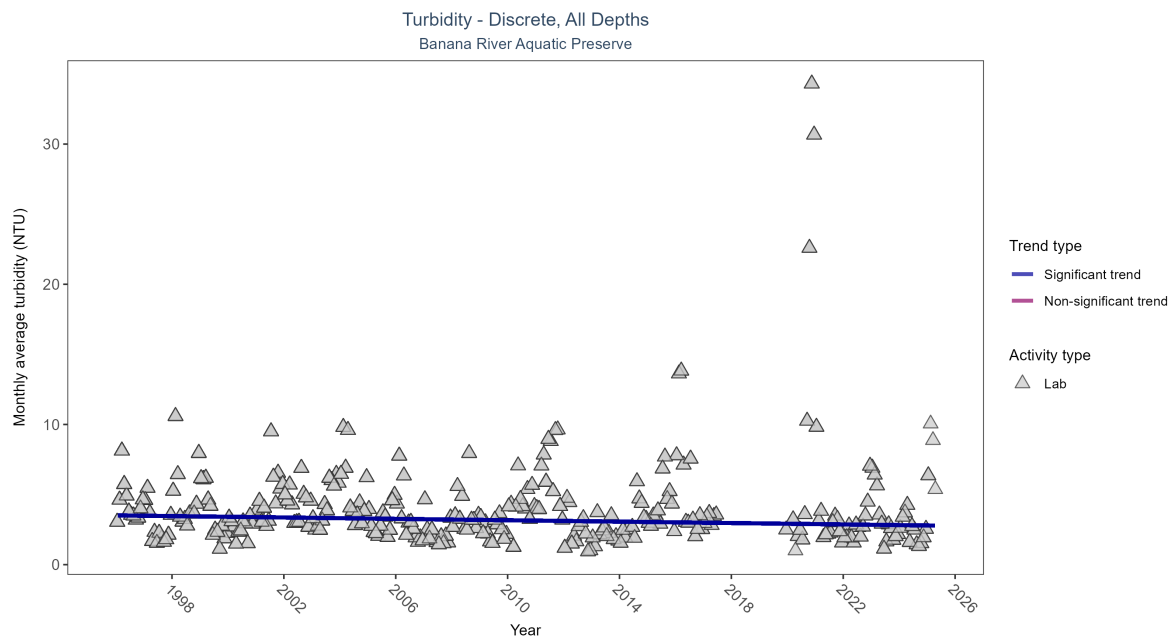


Figure 28: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 15: Seasonal Kendall-Tau Results for - Turbidity

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau    | Sen Intercept | Sen Slope | P     |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|--------|---------------|-----------|-------|
| Lab           | Significantly decreasing trend | 13558        | 29              | 1996 - 2025      | 3.13823             | -0.101 | 3.51387       | -0.02481  | 0.011 |

Monthly average turbidity decreased by 0.02 NTU per year, indicating an increase in water clarity.

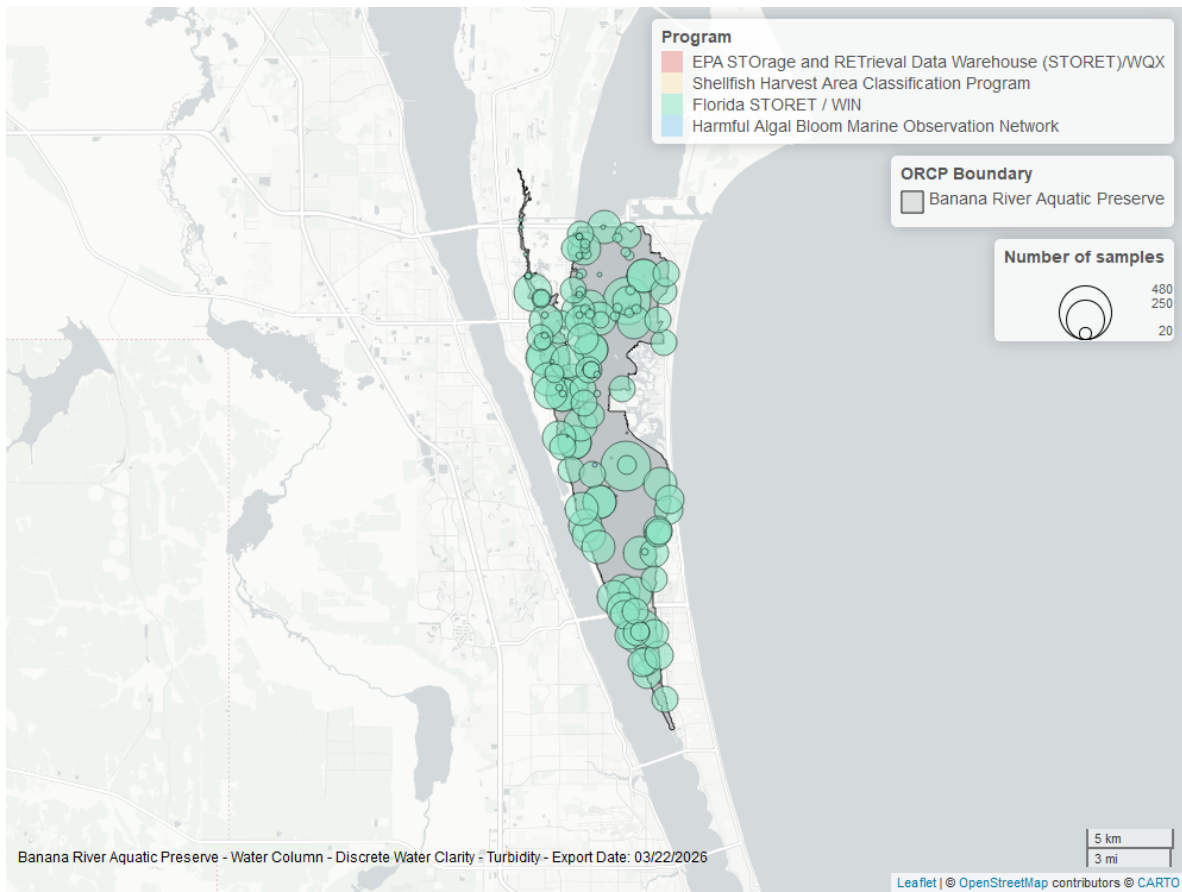


Figure 29: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Total Suspended Solids - Discrete

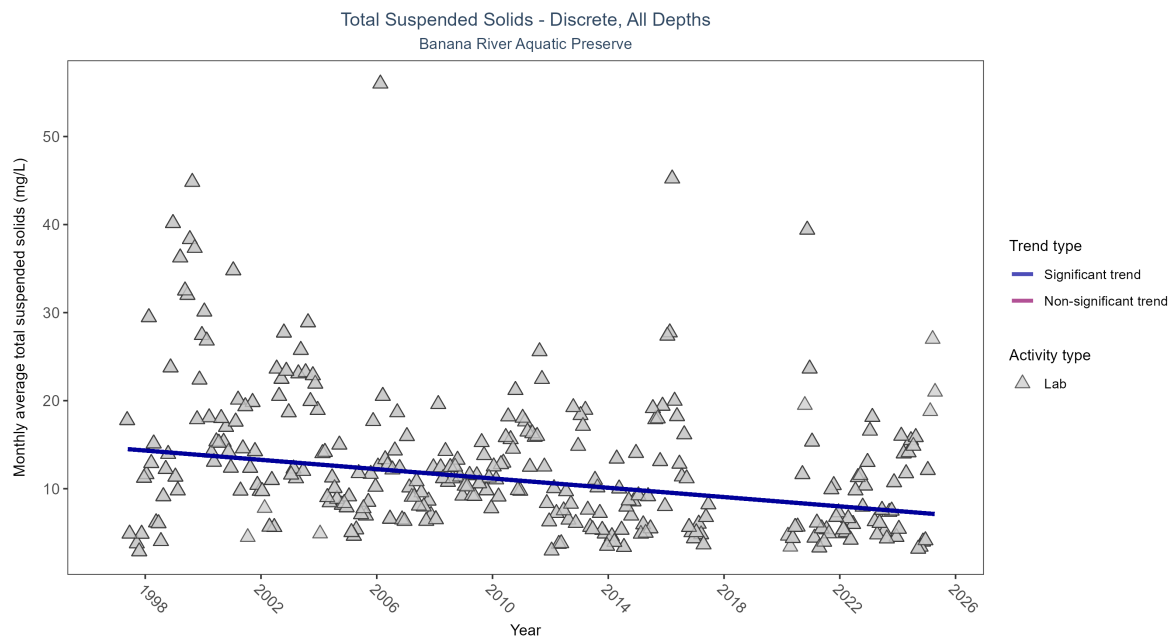


Figure 30: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Seasonal Kendall-Tau Results for - Total Suspended Solids

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|---|
| Lab           | Significantly decreasing trend | 2539         | 27              | 1997 - 2025      | 10                  | -0.22896 | 14.59995      | -0.26423  | 0 |

Monthly average total suspended solids decreased by 0.26 mg/L per year, indicating an increase in water clarity.

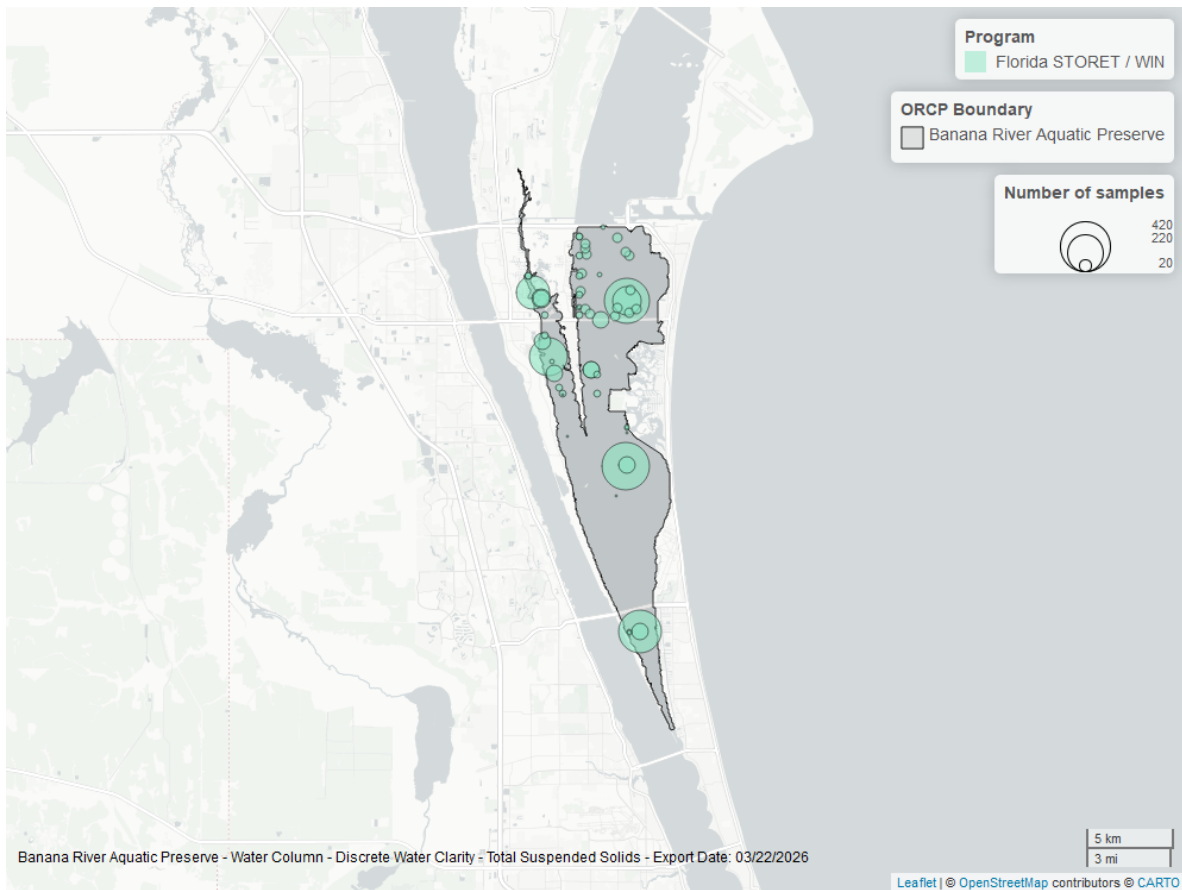


Figure 31: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Chlorophyll a, Uncorrected for Pheophytin - Discrete

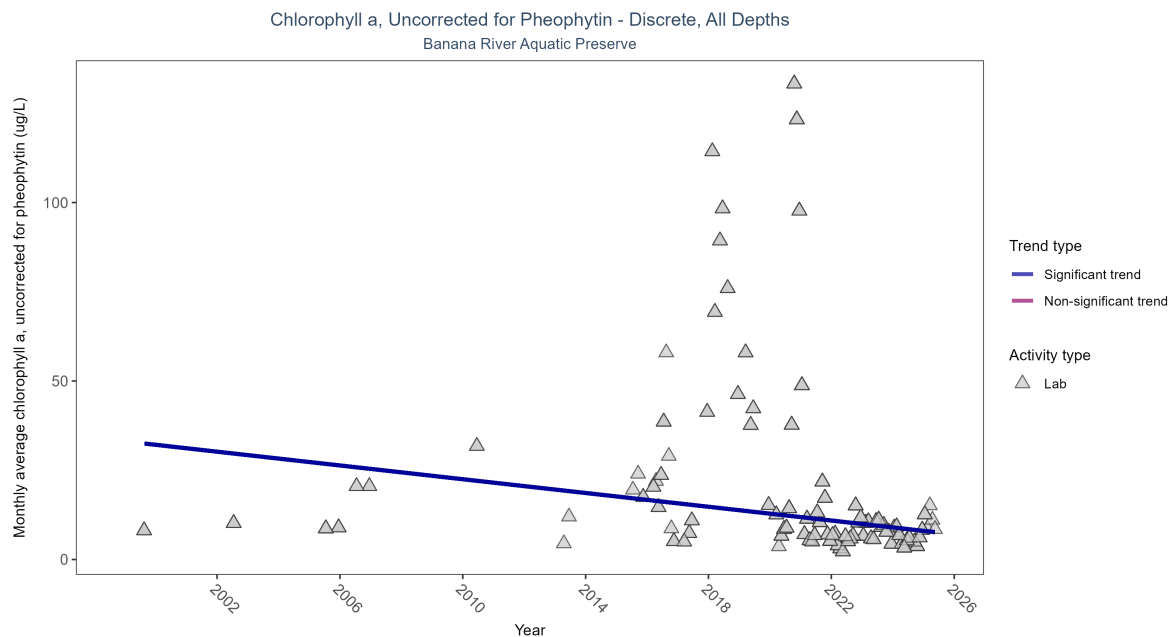


Figure 32: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|---|
| Lab           | Significantly decreasing trend | 519          | 17              | 1999 - 2025      | 7.60076             | -0.32728 | 33.10097      | -0.96434  | 0 |

Monthly average chlorophyll a, uncorrected for pheophytin, decreased by 0.96 µg/L per year, indicating an increase in water clarity.

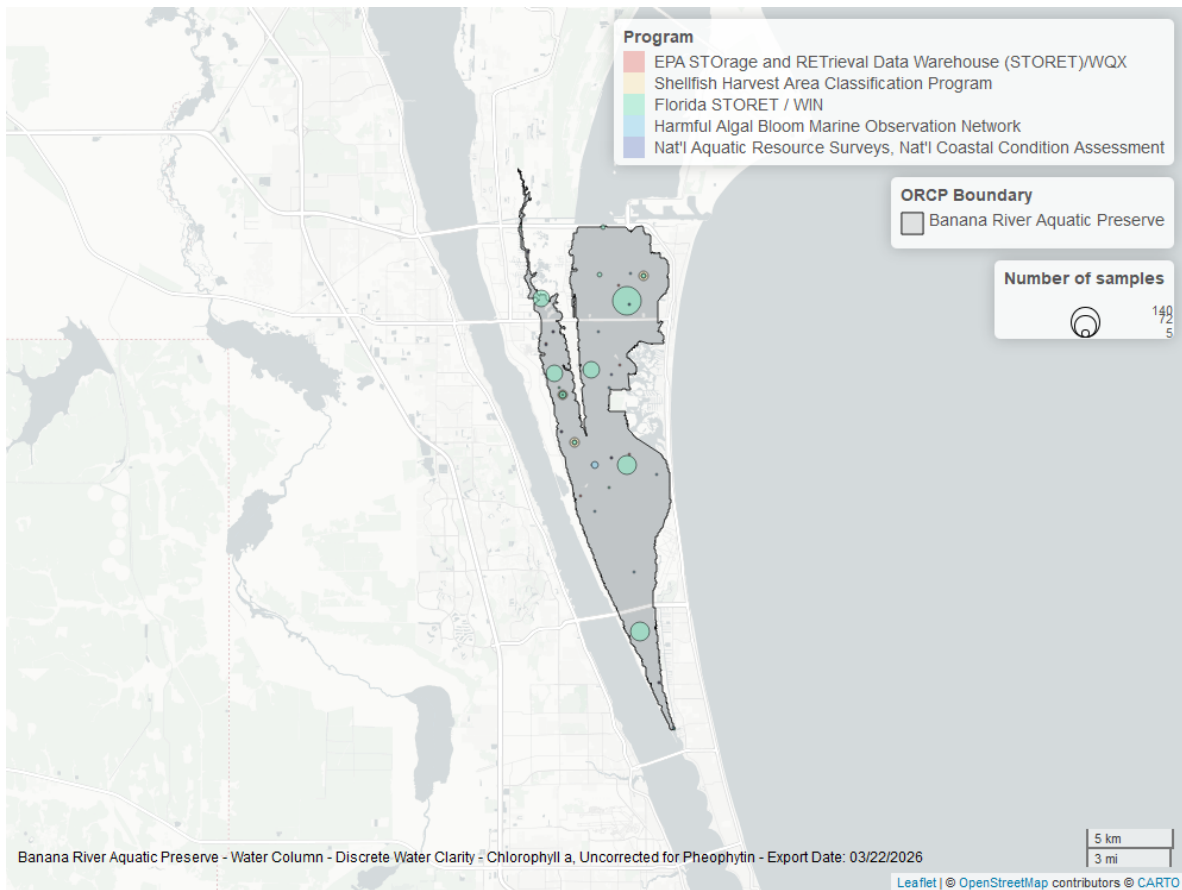


Figure 33: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Chlorophyll a, Corrected for Pheophytin - Discrete

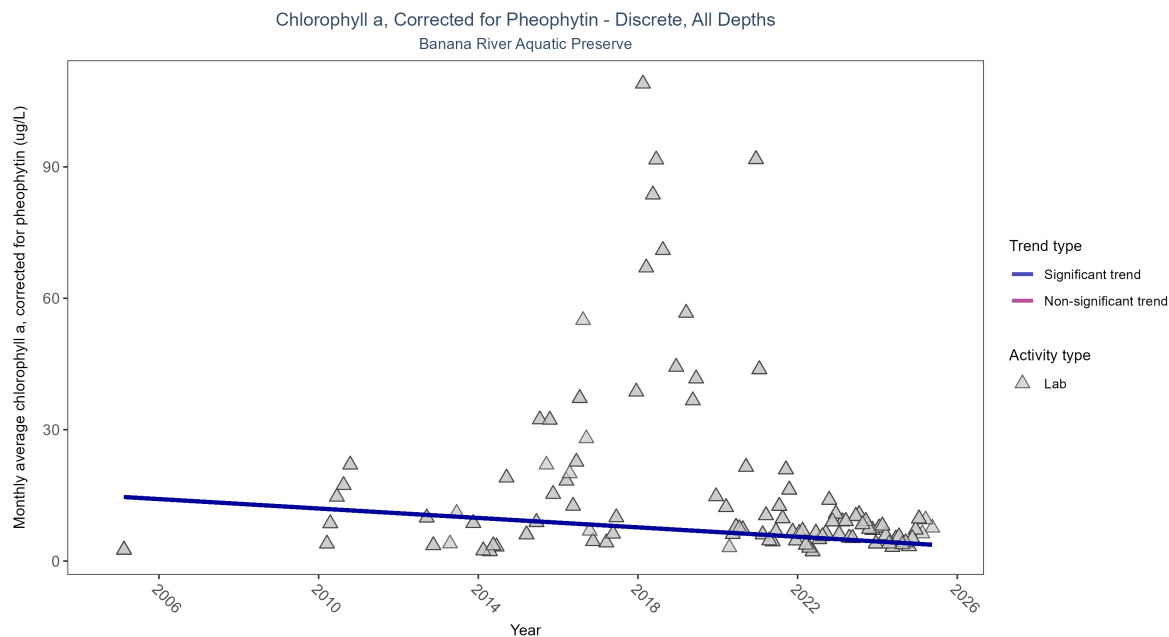


Figure 34: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 18: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P      |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|--------|
| Lab           | Significantly decreasing trend | 555          | 16              | 2005 - 2025      | 6.4614              | -0.2369 | 14.68847      | -0.53791  | 0.0046 |

Monthly average chlorophyll a, corrected for pheophytin, decreased by 0.54  $\mu\text{g/L}$  per year, indicating an increase in water clarity.



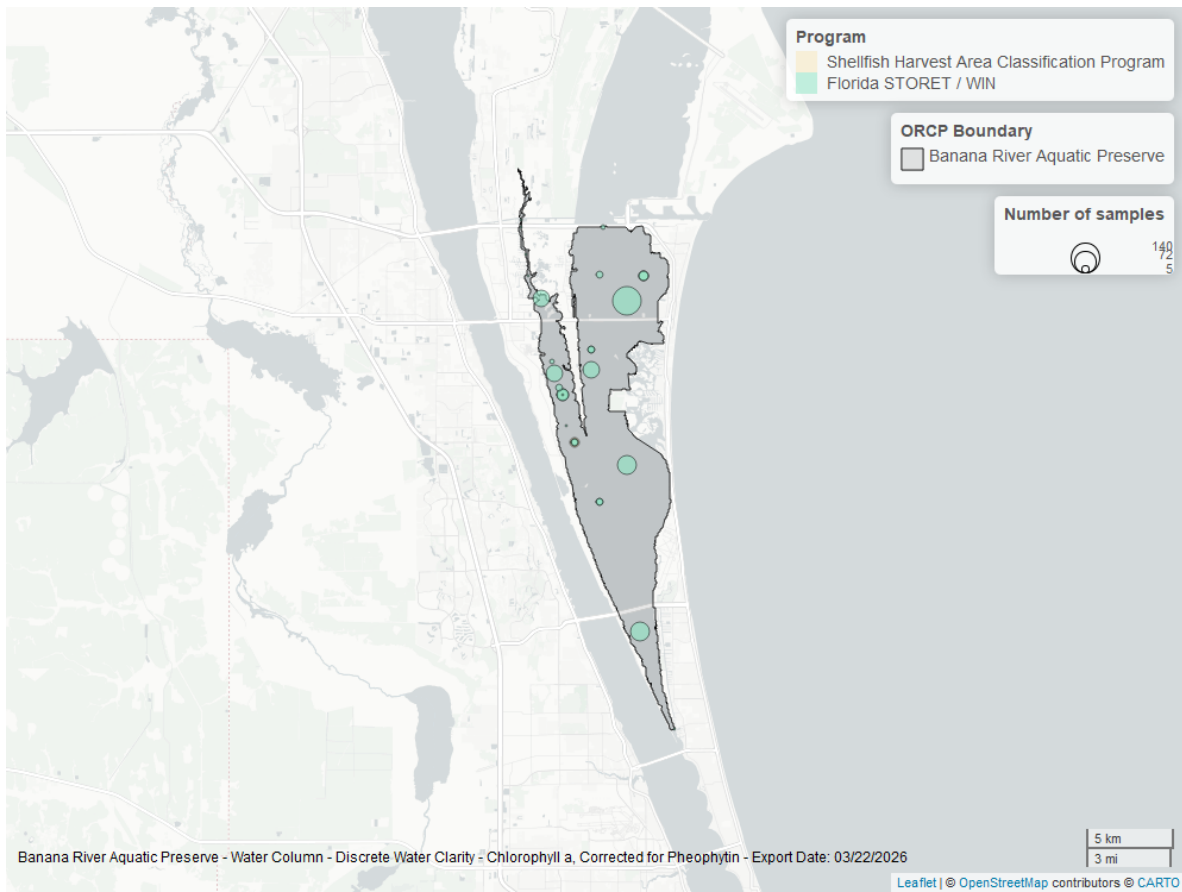


Figure 35: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Secchi Depth - Discrete

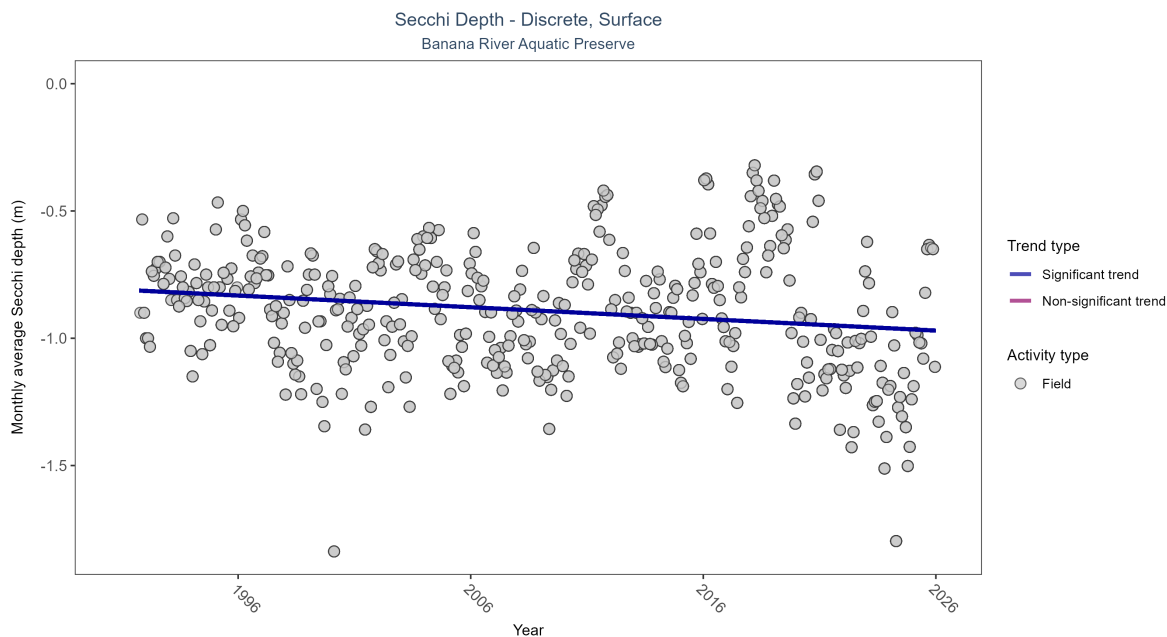


Figure 36: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 19: Seasonal Kendall-Tau Results for - Secchi Depth

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P      |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|--------|
| Field         | Significantly decreasing trend | 9257         | 35              | 1991 - 2025      | -0.9                | -0.11538 | -0.80935      | -0.00459  | 0.0009 |

Monthly average Secchi depth became deeper by less than 0.01 m per year, indicating an increase in water clarity.

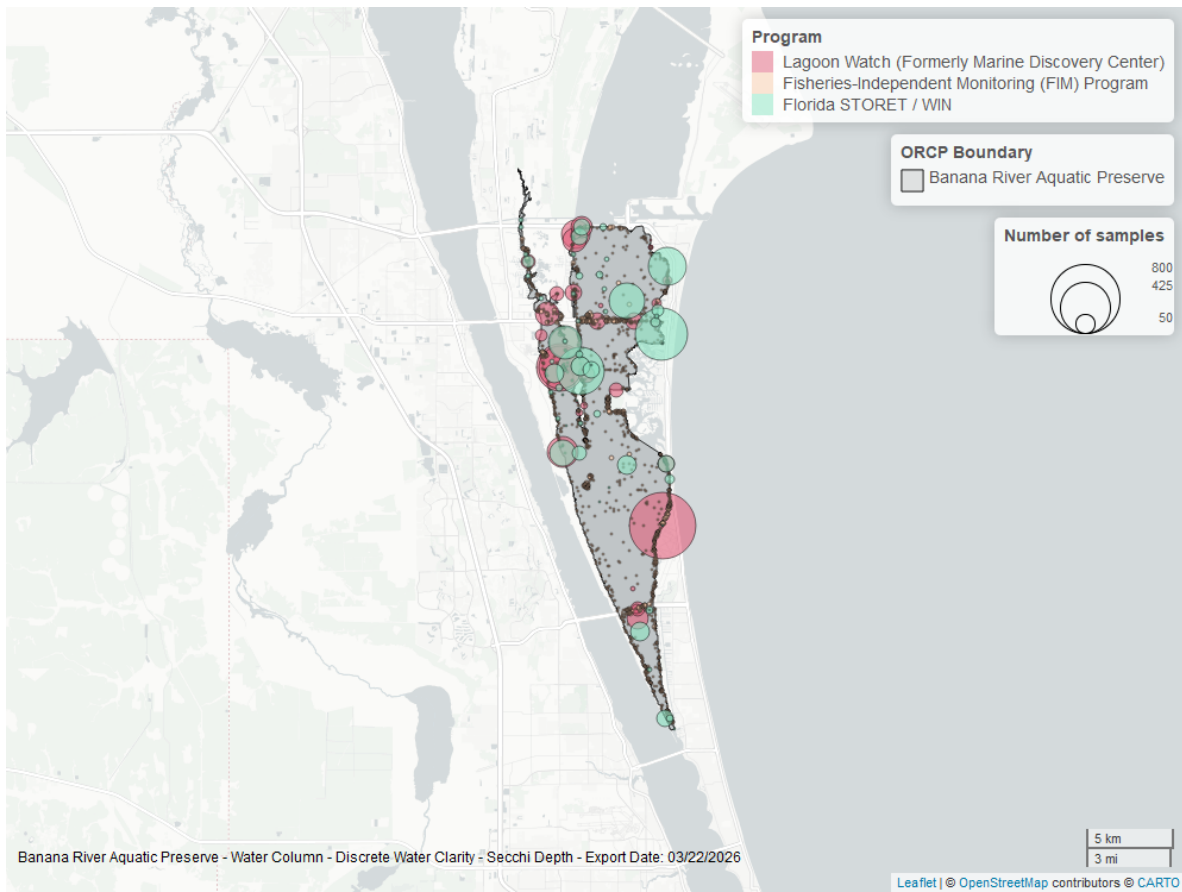


Figure 37: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Colored Dissolved Organic Matter - Discrete

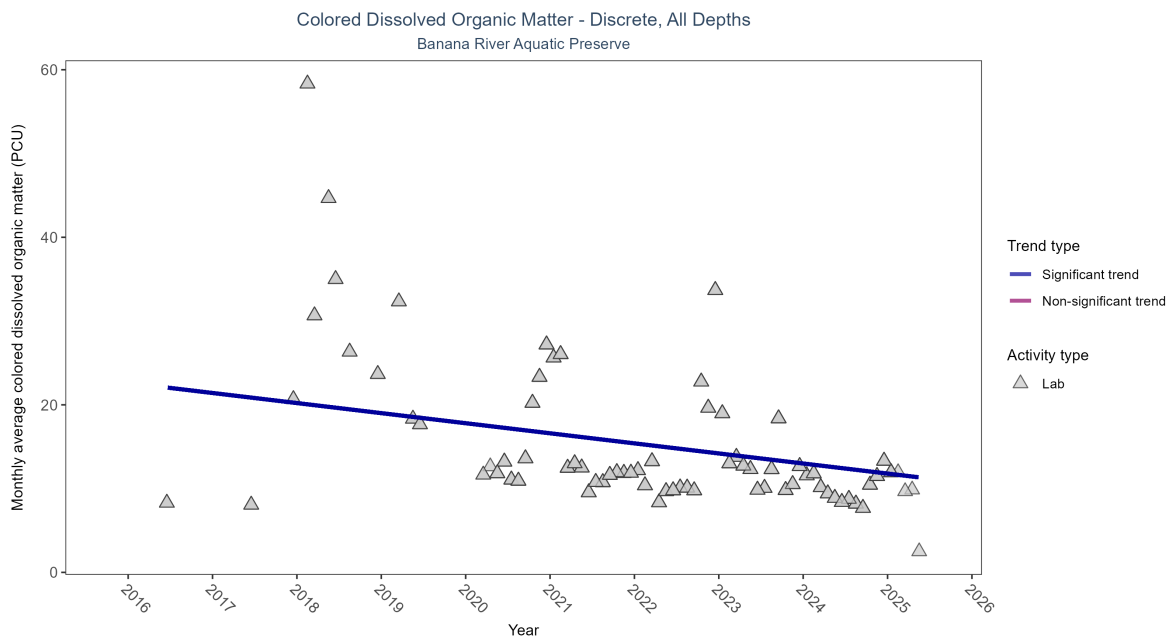


Figure 38: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 20: Seasonal Kendall-Tau Results for - Colored Dissolved Organic Matter

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|---|
| Lab           | Significantly decreasing trend | 456          | 10              | 2016 - 2025      | 11.33001            | -0.44959 | 22.61612      | -1.20252  | 0 |

Monthly average colored dissolved organic matter decreased by 1.2 PCU per year, indicating an increase in water clarity.

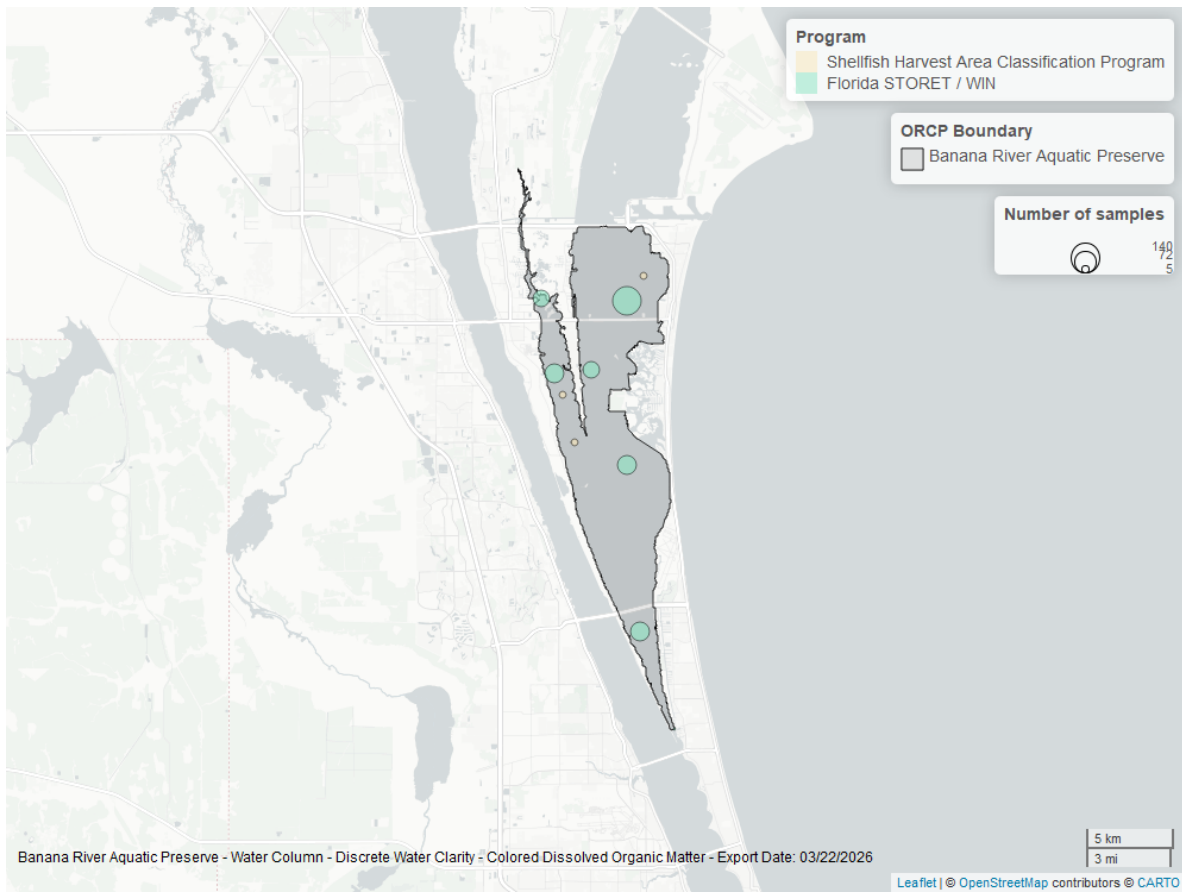


Figure 39: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.